Prediction Is Not Permission: Cross-Domain World Models Under Deterministic Runtime Authority

Architecture-Controlled and Incident-Motivated Evidence in Two Runtime Domains

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Technical Report v1.2 - 19 October 2026

Research artifact accompanying the FerrumOS world-model evidence lineage.

Abstract

World-model accuracy is not operational safety. This report separates six evidence objects: dynamics prediction, counterfactual response, warning quality, effective intervention, realized outcome, and independently enforced authority. An 18-run matched study in FerrumOS and Physical JEPA shows that model ranking depends on domain and episode composition. Frozen tests retain zero learned intervention on delayed hazards, an all-stop PyBullet policy, a Safety-Gymnasium result for which superiority over the planner was not established, and an output-value ablation that raises false warnings without resolving benefit. Three registered prediction studies follow. The v2 study reduces five-step error by 55.4% against OS persistence and 15.4% against same-data physical retraining. Studies v3 and v4 add 24 newly verified incident sources, ten abstract mechanism families per domain with three held out, and gradient-boosted residual corrections on the frozen v2 models. In v4, five-step error is 38.0% and 24.1% below the v2 physical model on the v2 population and on held-out mechanisms, and 20.8% below the v2 OS model on held-out mechanisms, with adjusted intervals excluding zero. The registered rule still fails: on the v2 OS population, where every episode reverses its drift unannounced, the five-step gain is 5.7%. The broader v3 failure is retained. A Dream-RSI-inspired evolving-world curriculum was 1.6% worse than its uniform counterpart. Warning gains are uneven: OS recall rises with false-positive rates up to 2.42%, and physical recall does not improve. Public narratives motivate local perturbations, not telemetry. Prediction gains do not establish controller benefit, physical safety, independent replication, or deployment eligibility; protected artifacts remain unchanged.

1. Introduction

An autonomous system can predict what may happen without possessing the authority to make it happen. That distinction is easy to state and easy to blur. A world model may forecast that a file operation will exhaust disk space, that a service restart will destabilize a workload, or that a commanded motion will approach an obstacle. The forecast may be useful even when imperfect. It does not follow that a low predicted risk should grant a capability, bypass confirmation, or energize an actuator.

This report asks two connected questions. First, when training conditions are controlled, is one common model family consistently better at multi-horizon prediction across an agentic operating system and a cyber-physical state space? Second, does the selected learned model add useful caution beyond a frozen deterministic authority boundary at a registered false-positive operating point? The first question concerns predictive modeling. The second concerns operational decision value. Treating

them as different estimands is the paper's central methodological choice.

The study extends two artifact-backed FerrumOS lineages: unprivileged action-conditioned OS forecasts before capability-gated effects, and a compact physical model evaluated with actuator authority disabled. It adds matched architectures, paired interventions, calibration, authority-disabled runtime tests, external-data intake, retained negative stress tests, a prospective Safety-Gymnasium evaluation [17], and a retrospective analysis of complete catalogs retained from a failed prospective output-value ablation. Section 8.7 then asks a distinct question: can incident-motivated development substantially improve prediction and warnings under a new frozen protocol without granting runtime authority? Section 8.8 asks whether a broader incident-motivated mechanism library, new model families, and a Dream-RSI-inspired curriculum improve on the models selected in Section 8.7, on that paper population and on mechanism families held out entirely from development.

1.1 Contributions

The primary contribution is an evaluation method, not a new JEPA objective. It predefines the same six evidence objects named in the abstract; specifies how final data are sealed; retains invalid, failed, and non-beneficial stages; and binds each claim to a machine-readable record. The empirical contributions are:

1. an 18-run matched small-model comparison in two distinct domains
2. paired temporal, calibration, and common-episode analyses that expose when predictive skill fails to become operational value
3. an authority factorization in which learning may only add caution
4. QEMU, recorded-sensor, PyBullet, and Safety-Gymnasium integrations labelled by evidence class
5. a prospective controller/shield benchmark with planner-relative uncertainty
6. a fresh risk-source attribution with multiplicity, paired-direction, leave-one-out, and common-proposal diagnostics
7. a retained-catalog output-value intervention preserving its failed prospective record
8. a separately registered incident-motivated study with significant prediction and warning gains, unchanged authority, and explicit simple-model and retraining controls; and
9. two further registered studies (v3 and v4) that add 24 verified incident sources, a novel-mechanism holdout, boosted residual world models, and a negative Dream-RSI-inspired ablation, with the failed v3 rule retained

1.2 Claim boundary

> *Cross-domain denotes application of the same evaluation and authority methodology to two domains. The models, state meanings, action spaces, labels, controllers, and datasets are not transferred between them. The temporal catalogs and PyBullet stress are locally designed software tests. Safety-Gymnasium supplies a third-party task and costs, but the adapters, privileged planner, correction, execution, and analysis are local. Physical evidence is replay and software simulation, not live HIL; FerrumOS evidence is disposable QEMU, not production use. No result establishes formal non-bypass, independent replication, human-contact safety, universal model superiority, or deployment safety, and no protected artifact is replaced.*

2. Related work and research position

2.1 Predictive representations and action-conditioned world models

Joint-embedding predictive architectures learn in representation space instead of reconstructing every observation detail. V-JEPA demonstrates feature prediction for video without pixel-level reconstruction [1], while V-JEPA 2 extends predictive representation learning toward action-conditioned planning and reports deployment on Franka robot arms [2]. DINO-WM shows that visual features can support world-model prediction across control tasks without task-specific visual encoders [3]. These results motivate predictive abstractions, but they do not imply that the same architecture will dominate in compact structured state spaces or that a prediction should carry authority.

The present study differs in scope. It does not propose a new large-scale pretraining objective. It compares small models using matched domain-specific data, seeds, parameter budgets, and update budgets at a runtime decision boundary, then measures whether offline ranking survives calibration and an operational threshold. FLOPs and wall time are not equalized. Its novelty is primarily systems and methodology: authority remains separately enforceable, and every evidence class is labelled by what it can and cannot support.

2.2 Agentic computer environments

OSWorld evaluates multimodal agents across real computer tasks and documents a large gap between human and automated performance [4]. Such benchmarks measure task execution in interactive environments. FerrumOS instead studies the narrower mediation point between a proposed canonical action and an operating-system effect. The study does not claim a general desktop-agent benchmark. It asks whether a predictive runtime can be exercised without granting state-changing authority and whether policy, capability, and confirmation remain independently testable.

2.3 Runtime assurance and safety filters

Simplex-style systems separate a high-performance controller from a trusted safety controller and use a decision module to switch when safety is threatened [5]. Neural Simplex and Black-Box Simplex extend this idea to learned components while retaining a recoverable or verified fallback boundary [6, 7]. NASA runtime-assurance work similarly treats monitoring and intervention as an architectural assurance layer rather than proof that an advanced controller is correct [8, 9]. In robot learning, SHIELD combines learned dynamics with a control-barrier-function layer and hardware evaluation [10]. Calibrated Predictive Safety makes calibrated risk and deterministic shielding central to a simulation study [11].

FerrumOS follows the same broad separation principle but targets heterogeneous authority. In the OS domain, a block, capability denial, confirmation request, and syscall validation are distinct decisions. In the physical domain, model inference, simulation command, and actuator delivery are distinct events. This paper therefore describes a monotone caution branch: learned output may increase intervention, but it cannot convert a forbidden action into an allowed one and cannot create actuator authority.

2.4 Calibration and proper scoring

Thresholded safety decisions depend on probability quality, not only rank order or mean rollout error. Reliability diagrams, expected calibration error, and the Brier score expose different properties of predictive confidence [12, 13]. ECE is bin-dependent and should not be treated as a complete guarantee. This study reports Brier, ECE, reliability bins, epistemic and aleatoric diagnostics, OOD distance, and risk-coverage curves. It also tests the actual frozen threshold. The distinction matters because both originally selected models respond directionally to interventions while adding no operational caution at that original threshold.

2.5 Improving the model versus improving experiment search

Dream-RSI reuses discovery histories to evaluate experiment-selection policies through replay, with decisions restricted to revealed prefixes [18]. It is not a drop-in optimizer for these world-model weights. The v2 study (Section 8.7) retains every development trial and selects by validation evidence, but implements neither Dream-RSI's replay-policy optimization nor a controlled RSI ablation; no v2 gain is attributed to RSI. The v3 study (Section 8.8) adapts the loop structure to training worlds and registers an equal-family, equal-seed uniform counterpart, so an RSI contribution is tested rather than assumed. TD-MPC2 motivates multi-step model training [19]. The new study borrows that general training principle, not TD-MPC2's reinforcement-learning algorithm, planning system, or claimed results. Gradient-boosted decision trees [28] enter v3 as learned corrections on top of the frozen v2-selected model, and per-horizon correction models act like a learned damped trend [29]. Fitted ridge regression and boosted trees remain eligible competitors: complexity is not an outcome.

3. Authority factorization

The system separates four questions: what is predicted, what policy permits, what authority is available, and what effect is independently observed. Let s_t be the captured state, a_t a normalized proposed action, M a learned transition model, R an exact deterministic predicate, C a capability and confirmation decision, and E an effect executor. A simplified gate is:

```
predicted_risk = M(s_t, a_t)
deterministic_block = R(s_t, a_t)
learned_block = predicted_risk >= frozen_threshold
gate_block = deterministic_block OR learned_block
execute = (NOT gate_block) AND C(a_t)
postcondition = independently_observe(E(a_t))
```

The Boolean union is monotone with respect to caution: a learned false-safe score cannot erase deterministic_block. Equally important, NOT gate_block is not an authorization token. The capability and confirmation branch remains necessary, and the effect must be observed separately. In the physical experiments, the executor is absent or actuator delivery is forced to zero.

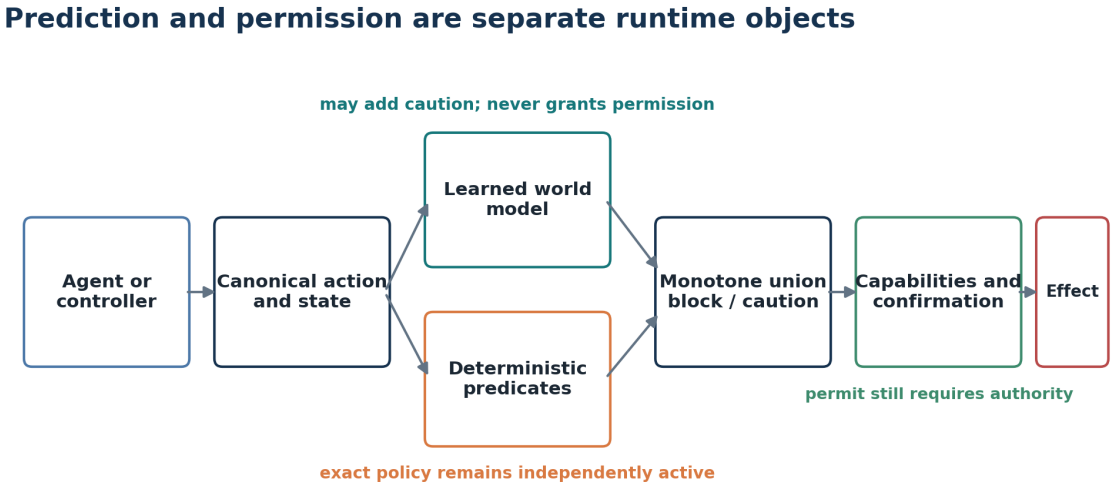


Figure 1. Authority factorization used in both domains. Prediction can add caution but cannot grant capabilities, bypass confirmation, or create physical actuator authority.

3.1 Why rollout horizon is not authority horizon

A multi-step model estimates a counterfactual trajectory under a specified action sequence. It does not mean that every predicted repetition was requested. This distinction previously mattered in FerrumOS, where repeated hypothetical application of a requested operation could create an artificial deterministic hazard. The repaired runtime applies exact deterministic effects only to covered requested actions while retaining multi-step learned output for telemetry and caution. The paper calls this request-bounded authority.

3.2 Failure semantics

The architecture is fail-closed only within stated boundaries. A missing or non-finite model cannot erase deterministic protection. A model that overestimates risk can deny useful work, so false positives and intervention rate are safety-relevant availability costs. A model that underestimates an unrepresented hazard may allow the proposal to proceed to capability and confirmation checks. Hazards absent from the state representation and deterministic predicates remain outside coverage. No empirical table converts these limitations into a formal guarantee.

3.3 Evidence classes

The study uses an evidence ladder rather than a single benchmark label. Sealed prediction supports comparative accuracy; deterministic catalogs support controlled estimands; QEMU shadow execution supports tested integration and no-dispatch claims; recorded testbed replay supports a bounded external-stream projection; and software physics supports simulated geometry and contact. Higher rungs add realism but do not retroactively turn lower-rung evidence into physical or independent assurance.

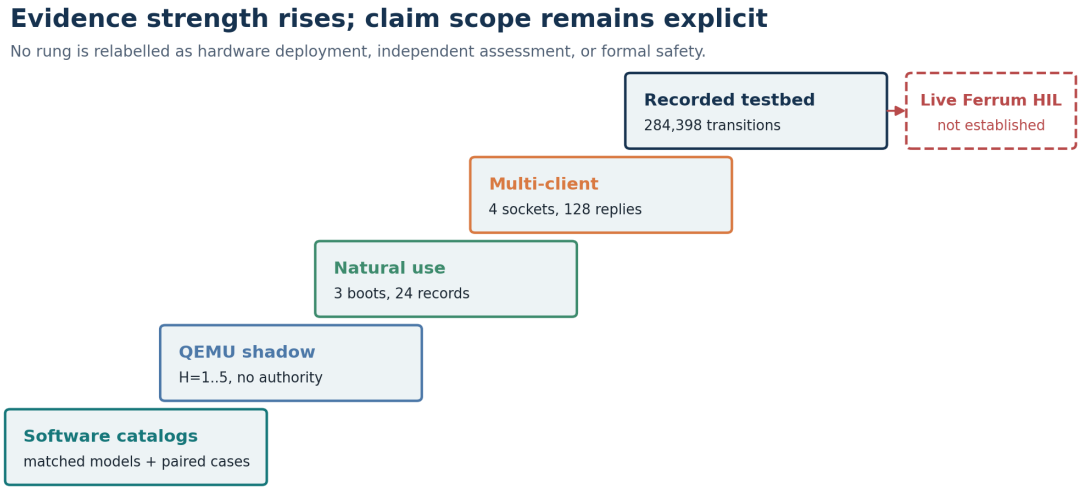


Figure 2. Evidence ladder and the strongest claim supported at each level. Higher levels add realism but do not erase the boundaries of lower-level measurements.

3.4 Authority threat model and enforcement tests

The threat model assumes a learned component may be wrong, missing, malformed, non-finite, stale, delayed, replayed, or paired with changed arguments or state. It also assumes callers may lack a capability or attempt to reach an effect through a separately enumerated interface. The tested contract is monotone blocking and independent authorization-not that blocking necessarily improves safety.

Table 1. Authority enforcement tests, their committed pass evidence, and what was executable during each test.

Component exercised	Test class	Committed pass evidence	Execution availability during test
FerrumOS learned-plus-rule gate	In-guest integration	world_model_failure_modes.json: pass	QEMU command path available; false-safe, missing, non-finite, and forbidden-coverage cases exercised
FerrumOS assistant mediation	In-guest system observation	world_model_natural_use_verification_v1.json: pass	Authorized reads available; writes required confirmation; deletes blocked
Signed neural permit protocol	Host unit	cross_domain_authority_test_inventory_v1.json: 9/9	Protocol library available; FerrumOS syscall path not exercised
Physical permit and disabled driver	Host unit	cross_domain_authority_test_inventory_v1.json: 128/128	Simulator/offline adapter only; physical actuator unavailable
Safety-Gym risk adapter runtime	Host integration	physical_jepa_safety_gymnasium_runtime_verification_v14.json: pass	Simulator command available; physical actuator unavailable

The mapped cases cover false-safe prediction, missing and non-finite artifacts, capability and permit denial, stale or replayed provenance, state revision, idempotency, and disabled physical delivery. They cover named protocol, daemon, and physical-runtime entry points, not every possible implementation path. In particular, physical permit unit tests do not establish FerrumOS syscall-path enforcement. Monotonicity means the learned branch cannot convert a deterministic *block* to *allow*; it does not mean an intervention is beneficial, as the PyBullet all-stop and planner-relative Safety-Gymnasium costs demonstrate.

4. Registered methods

4.1 Domains, representations, and partitions

FerrumOS uses 48-state and 57-action-feature vectors for canonical operating-system transitions. Physical JEPA uses 16-state and 10-action-feature vectors (seven action identities plus three continuous features). Both use a three-step causal history; short prefixes repeat the first state and prepend zero actions. They share metric code but not semantics. FerrumOS contains 13,712 training, 3,477 validation, and 3,249 final transition examples; its final set combines 1,969 base-test rows with four held-out incident sources. Physical contains 187,200 training, 45,440 validation, and 20,480 final transitions; its final set has eight held-out incident families. Selection programs cannot read final paths.

4.2 Matched architecture study

All methods predict the same normalized observable next-state delta, (next_state - current_state) / registered_scale, and a diagonal log variance. H-step rollout repeatedly adds the predicted delta in physical units while consuming the registered future action sequence. Normalized rollout error is the per-row mean absolute endpoint error divided componentwise by the same scale. Gaussian NLL and MAE therefore evaluate the same normalized target; NLL differs only by incorporating the predicted variance.

The direct MLP flattens three state-action pairs into Linear-GELU-Linear. The action-conditioned JEPA applies a linear state encoder, tanh, and a Linear-GELU-Linear latent predictor; a stop-gradient target encoder is updated with EMA coefficient 0.99, and training minimizes Gaussian NLL plus 0.25 times latent-target MSE. The GRU receives the same state-action sequence, uses its zero-initialized recurrent

state, and maps the final hidden state to mean and variance. The stop-gradient EMA target and observable-delta NLL are anti-collapse design choices, not a proof that collapse cannot occur. Selection checks prediction finiteness, but this matched study retains no explicit latent-variance or effective-rank collapse diagnostic.

Table 2. Matched architecture configurations: width, trainable parameters, and selected checkpoint updates per seed.

Domain	Method	Hidden / latent	Trainable parameters	Selected checkpoint updates for seeds 17 / 43 / 101
FerrumOS	Direct MLP	242 / -	99,800	1800 / 2000 / 2100
FerrumOS	Action-conditioned JEPA	211 / 64	99,748	2400 / 2400 / 2400
FerrumOS	GRU dynamics	125 / -	99,096	2400 / 2200 / 2200
Physical	Direct MLP	900 / -	99,932	2400 / 2200 / 1800
Physical	Action-conditioned JEPA	777 / 24	99,911	2400 / 2400 / 2400
Physical	GRU dynamics	164 / -	99,744	2400 / 2100 / 2400

Every run receives the same domain-specific split, seed set, batch size 512, maximum 2,400 AdamW updates, learning rate 0.001, weight decay 0.0001, and gradient clipping at 5. Lowest validation Gaussian NLL at 100-update checkpoints selects the saved state. The comparison is parameter- and update-matched, not strictly compute-controlled: FLOPs, wall-clock training time, and recurrent versus feed-forward execution cost were not equalized.

4.3 Aggregation, temporal causality, and uncertainty

Table 3 uses an ensemble prediction: the three seed checkpoints each predict, their state predictions are averaged, and the endpoint error is then computed. It is not the best seed or the mean of the three member errors. Appendix D (Table D1) reports every member result. Episode bootstrap intervals resample final episodes while holding the trained checkpoints fixed; they measure final-layout sampling uncertainty, not the variability of retraining the architecture on new seeds.

Paired temporal catalogs vary an intervention while sharing the initial state and random schedule. Directional accuracy asks whether the consequence ranking changes in the registered direction; ITE normalized MAE measures the magnitude difference. Multi-action rollouts use the actual registered action sequence rather than repeating one command. Uncertainty diagnostics include ensemble dispersion, learned variance, OOD distance, reliability bins, Brier score, ECE, and risk-coverage curves; none is an authority token.

4.4 Learned-contribution benchmark

For each domain, the lowest-validation-NLL architecture is frozen. Platt calibration and the intervention threshold are fitted on development data. The registered selection rule preserves zero false positives and produces a threshold of 0.99. Only after the model, calibration, and threshold are fixed does the generator create a 512-case final catalog containing 256 simulator-labelled dangerous cases and 256 matched safe cases.

Three paired arms are evaluated: rules only, learned only, and rules plus learned. Marginal learned hazard avoidance counts dangerous cases blocked by the learned branch but not by rules. Marginal safe intervention counts safe cases stopped only by the learned branch. Wilson intervals summarize binomial zero counts; a 5,000-pair bootstrap evaluates the marginal difference. The absent external-design manifest forces the label researcher-designed blinded deterministic-software benchmark.

4.5 Runtime and external evidence

FerrumOS shadow evaluation injects the research artifact only into disposable appliance-disk copies under QEMU/WHPX. Read-only preview follows the real ring-3 inference path but cannot call the execution method. A multi-client protocol uses four distinct WebSocket clients. A natural-use protocol operates three separately booted visible guests through ordinary assistant interactions and records only privacy-bounded event fields.

Physical evaluation has four separate components. Recorded HAI testbed streams are projected through previously frozen statistics and replayed with registered latency, jitter, noise, dropout, and combined faults. NVIDIA Anchor-Lab files are inspected for semantic compatibility before inference. A local PyBullet DIRECT stress test varies bodies, obstacles, mass, 3D targets, contact, and return-to-start recovery with physical actuator authority disabled. Finally, Safety-Gymnasium v1.0.0 supplies the externally maintained SafetyPointGoal1-v0 task and simulator costs. A registered adapter compares a naive local controller, a privileged deterministic grid planner, rules-only and learned-only shields, and their monotone union while keeping actuator authority zero.

4.6 Frozen risk-source attribution

The v2 attribution protocol holds SafetyPointGoal1-v0, the privileged planner, one-command tangent correction, 20-step oracle, Physical JEPA v5 artifact, episode limit, runtime lock, and seeds fixed. It varies only the warning pipeline: the full frozen v14 adapter; local lidar/action/goal/closeness/speed with JEPA weights zero; frozen JEPA clearance/velocity/progress outputs only; or current maximum hazard closeness only. Alternative logistic adapters are fitted on seeds 4000-4095 and select thresholds on 4096-4127 by matching the full adapter's validation FPR, then minimizing false negatives, then choosing the higher threshold. Final seeds 8000-8127 are opened once.

The first registered attribution attempt on seeds 7000-7127 failed after the baseline and full arm because the alternative JSON schema omitted a runner-required threshold alias. Its partial catalog is retained and excluded. The v2 repair adds only that alias, preserves every numeric adapter field and threshold, commits the repair before final access, and uses a new seed range.

4.7 Retained-catalog output-value intervention

The output-ablation v1 protocol fixed SafetyPointGoal1-v0, the full v14 risk adapter, threshold 0.72, tangent correction, final seeds 9000-9127, two paired arms, all estimands, and bootstrap seeds before either catalog existed. The observed arm passes the three frozen Physical JEPA v5 outputs into the unchanged adapter. The masked arm computes the same outputs but replaces raw adapter features 22-24 with their preregistered development means immediately before scoring. A 10,000-resample paired episode bootstrap estimates observed-minus-masked changes. Pointwise 95% intervals are accompanied by a post-hoc Bonferroni sensitivity controlling 5% familywise error across the seven endpoints displayed in Section 8.6, yielding 99.29% intervals. A stricter audit across all 11 registered paired outputs checks whether the displayed-endpoint family changes the survivor set. This is a direct closed-loop intervention on output values within one fixed software pipeline; it is not an architecture replacement or an architecture-wide causal claim.

Both 128-episode catalogs completed during the prospectively specified v1 execution before a post-run protected-file check detected that another task had rebuilt the v1.1 PDF. The design and simulator execution were prospective, but the v1 result remains failed and ineligible because that final integrity gate did not pass. Recovery protocol v2 reused the retained catalogs without simulator execution and wrote the complete retrospective reportable analysis. Verification amendment v3 corrects checkout-dependent newline hashing, validates the historical v1.1 PDF blob at the recovery registration commit, and passes 51/51 checks. It changes no estimand, metric, result, seed, model, adapter, or authority state and does not reclassify the v1 result as successful.

4.8 Frozen gates, hashes, and non-promotion

Validation means checking an existing record against its schema, hashes, and gates. Recompute means deriving metrics again from committed rows or episode summaries without simulator execution. Rerun means executing the simulator or QEMU workload again. The verifiers distinguish these operations. SHA-256 establishes byte identity of named files, not scientific correctness; program logs establish only the events instrumented by those programs; and a verifier may share assumptions with its producer. Git commit timestamps provide externally hosted chronology after push, not proof that equivalent data were never observed elsewhere. Passing any study verifier does not imply deployment eligibility. Protected FerrumOS and Physical JEPA deployment digests must remain unchanged, and the study explicitly sets promotion_eligible=false.

5. Architecture-controlled results

5.1 Domain-dependent ranking

Table 3 reports the error of the three-seed ensemble prediction defined in Section 4.3. Physical JEPA wins at every horizon. FerrumOS does not reproduce that ranking: the GRU is best at H=1 and H=3, while the JEPA is best at H=5. The registered pairwise intervals for the stated leaders exclude zero, conditional on the fixed trained checkpoints; Appendix D shows all seed members.

Table 3. Registered matched rollout error of each three-seed ensemble (normalized MAE; lower is better).

Domain	Method	H=1	H=3	H=5
FerrumOS	Direct MLP	0.004751	0.008431	0.007256
FerrumOS	Action-conditioned JEPA	0.009888	0.012356	**0.005494**
FerrumOS	GRU dynamics	**0.004609**	**0.007917**	0.006514
Physical	Direct MLP	0.007188	0.016836	0.026362
Physical	Action-conditioned JEPA	**0.002476**	**0.006680**	**0.010465**
Physical	GRU dynamics	0.010659	0.026231	0.039542

The physical MLP-minus-JEPA H=3 difference is 0.010156 with 95% interval [0.009829, 0.010483]. In FerrumOS, GRU-minus-JEPA at H=3 is -0.004439 [-0.005735, -0.003203], so the sign correctly favors the GRU. At H=5, JEPA-minus-GRU is -0.001020 [-0.001165, -0.000865]. These registered, horizon-specific contrasts are statistically separated under the stated resampling procedure, conditional on their eligible episode populations.

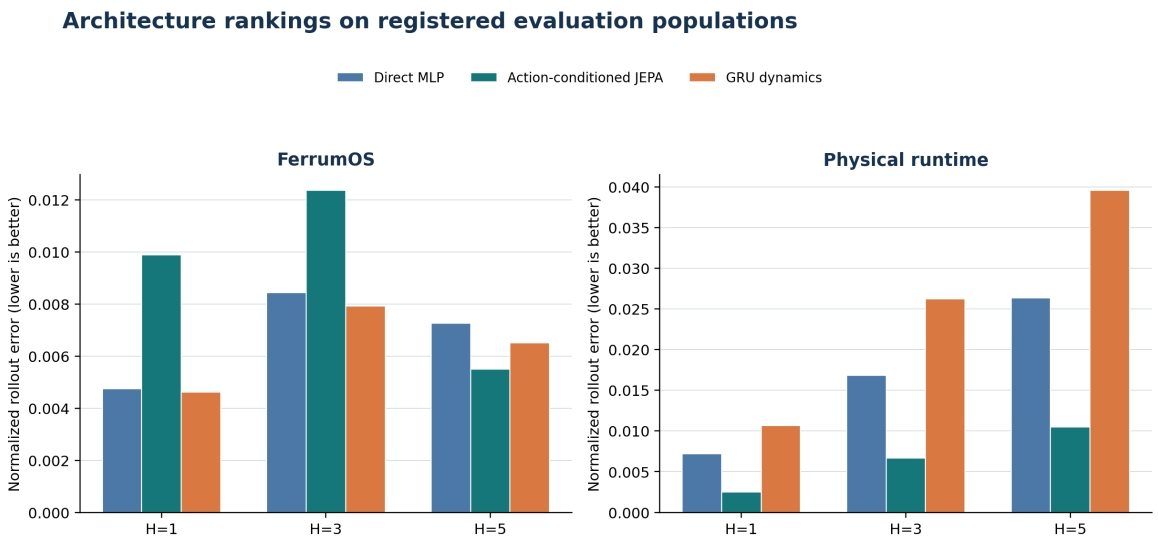


Figure 3. Registered matched rollout errors for three model families. The FerrumOS horizon populations differ; the common-episode sensitivity is reported separately.

5.2 Interpretation

The result closes data, parameter-count, and update-budget confounds that affected historical comparisons, but not FLOP or training-time differences. In the registered table, the tested three-member FerrumOS GRU ensemble leads at H=1 and H=3; this does not describe the mean individual checkpoint at H=1, where MLP is lower by only 0.000008. H=3 averages 318 eligible episodes, whereas H=5 averages 238 and excludes a source whose sequences are too short. Thus the registered ranking reversal cannot be isolated as a horizon effect: episode composition changes it. The original table and model selection remain unchanged.

Table 4 reports the complete post-hoc common-episode comparison. All estimates are three-member ensemble normalized errors on the same 238 FerrumOS episodes; lower is better. Contrast intervals are paired, unadjusted 10,000-resample 95% bootstrap intervals conditional on the fixed checkpoints and this episode population.

Table 4. Post-hoc common-episode comparison on the same 238 FerrumOS episodes, with paired 95% bootstrap intervals.

Horizon	Direct MLP	JEPA	GRU	MLP - JEPA, paired 95%	JEPA - GRU, paired 95%
H=1	0.002343	**0.001018**	0.002401	+0.001325 [0.001295, 0.001356]	-0.001383 [-0.001408, -0.001358]
H=3	0.006709	**0.005028**	0.006147	+0.001681 [0.001614, 0.001746]	-0.001119 [-0.001218, -0.001016]
H=5	0.007256	**0.005494**	0.006514	+0.001763 [0.001646, 0.001874]	-0.001020 [-0.001164, -0.000871]

JEPA has the lowest common-episode estimate at every horizon, and both displayed paired contrasts exclude zero. This post-hoc table changes the interpretation of the registered horizon-specific ranking; it does not replace the registered result or model selection.

The matched study also changes how the two earlier reports should be read. The Physical JEPA result can now be described as an architecture-controlled rollout advantage in its registered domain. The FerrumOS lineage cannot claim that a JEPA is the strongest general small dynamics model. Its stronger contribution is the mediation architecture, evidence discipline, and the request-bounded separation between prediction and authority.

6. Causal sensitivity, calibration, and operational value

6.1 Paired temporal results

The selected FerrumOS GRU produces the correct counterfactual direction in 100.00% of pairs, ITE normalized MAE 0.002571, H=3 multi-action error 0.044558, and H=5 error 0.046720. The selected Physical JEPA reaches 93.36% directional accuracy, ITE normalized MAE 0.016819, H=3 error 0.026772, and H=5 error 0.046361.

Table 5. Paired temporal results for the originally selected models.

Domain	Selected model	Direction	ITE MAE	H=3 multi	H=5 multi
FerrumOS	GRU dynamics	100.00%	0.002571	0.044558	0.046720
Physical	JEPA	93.36%	0.016819	0.026772	0.046361

Directional sensitivity establishes that the models do not merely reproduce a static current-state score. It does not establish calibrated hazard probability, adequate recall, or useful intervention. Multi-action errors are also substantially larger than the matched single-policy H=3 values in Table 3, which is consistent with a harder shifted temporal task.

6.2 Calibration under shift

FerrumOS Brier and ECE are 0.249943 and 0.238143. Physical Brier and ECE are 0.229970 and 0.292887. Because each final catalog is balanced, the constant-prevalence predictor has Brier 0.25; FerrumOS is effectively at that baseline and Physical improves it modestly. ECE remains bin-dependent, and the registered risk-coverage curves are non-monotonic. These scores describe the shifted catalog and do not certify an error ordering or an operational probability.

The frozen 0.99 threshold was selected on 256 development cases per domain to match the rules-only FPR of zero, breaking ties by lower FNR and then higher threshold. Zero observed development FPs do not guarantee zero population FPR. No threshold is retuned after final access.

6.3 Primary operational result

On each 512-case balanced final catalog, rules-only, learned-only, and union record TP=0, FP=0, TN=256, and FN=256 at threshold 0.99.

Table 6. Primary operational result on the 512-case final catalogs at the frozen threshold 0.99.

Domain	Arm	TP	FP	TN	FN	Intervention
FerrumOS	Rules only	0	0	256	256	0%
FerrumOS	Learned only	0	0	256	256	0%
FerrumOS	Rules + learned	0	0	256	256	0%
Physical	Rules only	0	0	256	256	0%
Physical	Learned only	0	0	256	256	0%
Physical	Rules + learned	0	0	256	256	0%

For each domain, the learned branch produces 0/256 marginal interventions on dangerous cases and 0/256 on safe cases. The two-sided Wilson 95% upper bound for either zero-observed marginal intervention rate is 1.478%. That quantity is not a miss-rate bound: the observed false-negative rate is 256/256, with an approximate two-sided Wilson interval of [98.52%, 100.00%]. The 5,000-pair bootstrap interval for the observed marginal difference is exactly [0, 0] because every paired value is zero. All eight final families are delayed, coupled, masked, or exogenous: danger materializes later, while the deterministic predicates inspect current state and action. rule_block is false in all 1,024 records by construction of this estimand, so rules-only recall is not a general rule-quality estimate. The

learned branch likewise does not extend the authority boundary at the zero-FP-calibrated threshold. This is a thresholded latent-hazard coverage negative, and no final-set retuning or promotion follows.

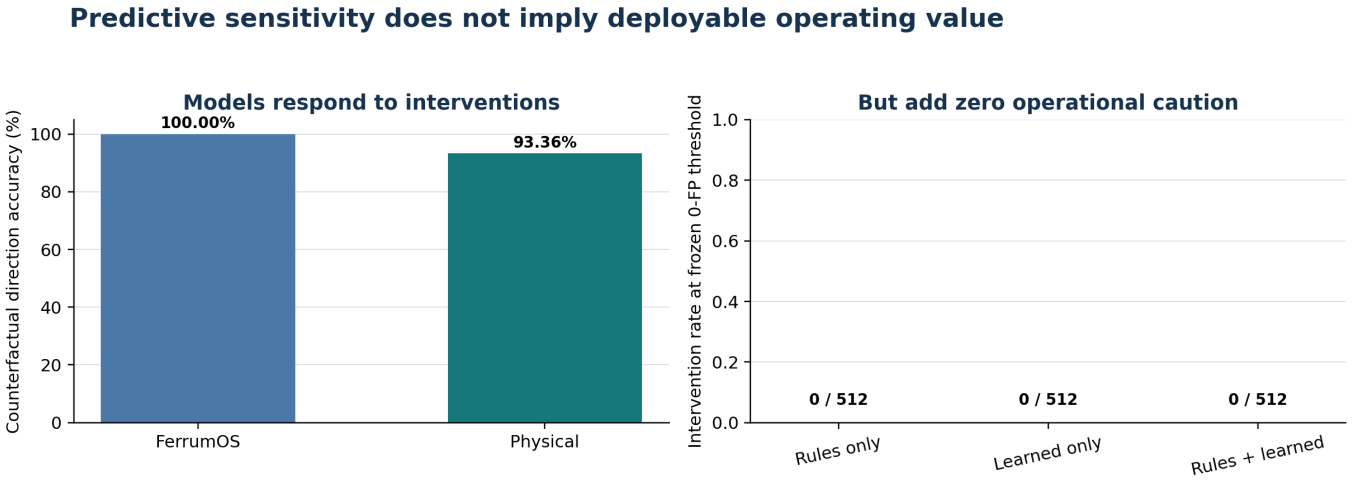


Figure 4. Predictive and causal evidence do not become operational learned value at the frozen threshold. The gap is measured rather than hidden by final-set retuning.

7. FerrumOS authority-separated runtime evidence

7.1 Lineage and measurement boundaries

Three original model objects must not be conflated. The architecture study trains approximately 100k-parameter research models solely for matched offline comparison. The QEMU runtime loads the earlier 193,229-parameter FerrumOS v3.4 encoder-plus-transition lineage. Physical experiments use the distinct frozen Physical JEPA v5 binary. The later incident-motivated candidates form a fourth, separately archived research lineage. Neither architecture-study nor incident-motivated checkpoints are substituted into either runtime.

Table 7. Runtime evidence objects, their clocks, and the timing claim each supports.

Evidence object	Model object	Clock and workload	Supported timing claim
Guest horizon probe	FerrumOS v3.4, one preview at a time	1 kHz guest timer; 200 previews per horizon	3 ms guest-reported p99 in that probe
Four-client saturation	Same serial QEMU daemon	Host WebSocket wall clock; 128 responses	17.545 s batch wall time, about 7.30 responses/s; client median 8.45-8.83 s and p95 16.09-16.46 s
Natural-use sessions	Same mediation lineage	Guest cycle counters without calibrated frequency	Ordering and bounded counts only, not milliseconds
HAI NumPy batch	Physical JEPA v5 on host arrays	Host batch timing, 4096 rows	Microseconds per row, not control-loop latency

These clocks and workloads are not directly comparable. The study has no synchronized decomposition of model compute, guest scheduling, serialization queueing, transport, and polling. “Serialized inference” is therefore a supported mechanism-level explanation for saturation, not a complete latency attribution.

7.2 Shadow integration and contention

The v3.4 model is injected only into disposable appliance-disk copies and loaded by the actual ring-3 preview gate. The one-at-a-time probe reports 3 ms p99 at H=1 through H=5, zero retained heap growth, and 96/96 correlated command-class responses. The preview emits no execution record and the source disk remains byte-identical. Under four-client saturation, all 128 responses return without leakage, Jain fairness is 0.999937, disconnect isolation succeeds, and a replacement client completes. Execution RPC is unavailable with error -32601, and execution records and physical deliveries remain zero.

7.3 Visible natural-use sessions

Three separately booted disposable guests produce 24 privacy-bounded rows: 18 reads execute, three writes await operator confirmation, and three deletes are blocked. Telemetry excludes prompts, arguments, paths, provider/model identifiers, and output text. This is short researcher-operated QEMU evidence, not production telemetry or independent user testing.

7.4 Runtime interpretation

The supported claim is narrow: a learned preview runs inside the tested unprivileged mediation path while capability, confirmation, revision checks, and execution remain separate. The source tests in Section 3.4 exercise false-safe prediction, malformed artifacts, stale/replayed messages, capability denial, and actuator-disabled delivery. This establishes empirical enforcement for enumerated paths-not a formal proof of non-bypass and not evidence that every block improves safety.

8. Physical evidence and retained negatives

8.1 Externally authored case intake

The external intake freezes Microsoft Azure VM-noise revision `207bed67dd10090b28ad4f745b2cfd41a11aace4` as an OS workload taxonomy and NVIDIA Anchor-Lab revision `647edd5787cd764cdc041103ad282dc59214d919` as physical telemetry. Azure families cover file, process, thread, program launch, CPU, memory, random I/O, and Redis contention. They inform the natural-use and contention mix but are not projected into FerrumOS vectors or treated as FerrumOS labels.

Six Anchor-Lab Parquet files contain 2,925,558 finite timestamped rows, including three publisher-designated held-out SO-101 trials and three H1 elbow-teststand conditions. The files contain command or target and measured-state telemetry but no safety or contact labels. Their joint, actuator, and temperature semantics do not match Physical JEPA v5's 16-state navigation and maintenance representation or seven-action ontology. No projection, inference, or actuator delivery was performed. The incompatibility is a useful negative intake result: it prevents a large external row count from being misrepresented as direct model validation.

8.2 Recorded HAI sensor replay

The frozen Physical JEPA v5 artifact is evaluated over 284,398 one-second transitions from HAI 23.05 test1 and test2 using previously frozen projection statistics. Six conditions apply clean replay, two-second latency, zero-to-three-second jitter, 0.02 sensor noise, 5% hold-last dropout, and a combined fault with proxy-command feature saturation.

Table 8. Recorded HAI replay under registered observation faults (Physical JEPA v5, actuator delivery disabled).

Condition	H=1 MAE	Delta	AUROC	Event recall
Clean	0.046627	0.000000	0.6115	38.46%
Latency 2 s	0.050421	0.003794	0.6040	48.08%
Jitter 0-3 s	0.049488	0.002861	0.6039	44.23%
Noise 0.02	0.055194	0.008567	0.5937	46.15%
Dropout 5%	0.046735	0.000108	0.6105	38.46%
Combined	0.059456	0.012829	0.6043	50.00%

Host NumPy batch inference measured 2.239 microseconds median and 3.303 microseconds p99 per row. These are host software measurements, not physical control-loop timing. Across 472 injected observation-fault windows, the first clean row after the injection ended fell below 1.25 times clean p95 error. That zero-second observation restoration is not a physical recovery-time claim. Actuator delivery attempts and deliveries were both zero.

The projection is fixed from 698,400 HAI train1-train3 rows and maps only eight available source channels into the 16-state representation; missing state dimensions retain registered constants. Test1 labels align at exact second resolution; test2 labels align positionally only after matching the source file's minute precision, with equal row counts and no shifting, interpolation, nearest-neighbor matching, fill, or imputation. Event labels and proxy commands are descriptive anomaly diagnostics, not equivalent to the model's maintenance/action ontology. The replay is therefore bounded external-stream compatibility evidence, not live Ferrum HIL, end-to-end safety validation, physical timing, actuator dynamics, contact recovery, or an independently tested emergency stop.

8.3 Multi-embodiment 3D stress

A separate PyBullet DIRECT protocol varies three bodies, three obstacle geometries, mass, three-dimensional targets, contact, and a one-second return-to-start recovery. All 288 cases are stopped by the union policy. Task completion is 0%, with Wilson 95% upper bound 1.32%. The unshielded arm contacts an obstacle in 205 of 288 cases. The union arm records no contacts only because it intervenes in 288 of 288 cases.

The learned branch contributes 23 interventions not produced by the rule; 13 coincide with an unshielded contact. Those cases cannot be credited as useful learned collision avoidance because the policy has no selectivity and completes no task. Simulated return-to-start recovery succeeds in 181 of 205 contact cases, or 88.29%, with Wilson interval [83.17%, 92.01%]. This locally designed software-physics test adds geometry and contact coverage while failing the completion-versus-intervention objective.

Table 9. PyBullet DIRECT multi-embodiment stress outcomes (288 cases).

3D outcome	Count	Rate or interval
Union interventions	288 / 288	100.00%
Completed tasks	0 / 288	0%; upper 95% 1.32%
Unshielded contacts	205 / 288	71.18%
Learned-only interventions	23 / 288	7.99%
Learned-only and unshielded contact	13 / 288	4.51%
Simulated recovery	181 / 205	88.29% [83.17%, 92.01%]

No shielded contact was observed in this run; this does not establish a zero underlying collision probability. More fundamentally, avoiding contact by stopping every case is not useful autonomy. Retaining the negative prevents an embodiment-diversity checkbox from being mistaken for deployment progress.

8.4 Prospective Safety-Gymnasium controller and shield benchmark

The v14 amendment freezes Safety-Gymnasium v1.0.0, Gymnasium 0.28.1, MuJoCo 2.3.3, the installed simulator-source digest, the protected Physical JEPA v5 digest, a deterministic risk adapter fitted on opened seeds 4000-4095, candidate choice on opened seeds 4096-4127, and untouched final seeds 6000-6127. The 20-step oracle rolls out the nominal receding-horizon controller from synchronized simulator state; it does not repeat the current command for 20 steps. Warning recall and warning FPR evaluate the detector, whereas intervention rate counts only commands that actually change. The external project supplies the task, layouts, observations, goal condition, and hazard costs [17]. This study supplies the adapter, privileged planner, tangent shield, execution, and analysis. Execution is local, actuator authority is disabled, and no independent replication is claimed.

Table 10. Prospective Safety-Gymnasium final arms on untouched seeds 6000-6127.

Final arm	Completion	Effective intervention	Warning recall	Warning FPR	Effective-action recall	Hazard-cost events
Naive unshielded	100.00%	0.00%	-	-	0.00%	542
Planner unshielded	94.53%	0.00%	-	-	0.00%	**70**
Planner + rules	94.53%	0.00%	0.00%	0.00%	0.00%	70
Planner + learned	**96.09%**	1.86%	**95.62%**	3.24%	55.00%	84
Planner + rules + learned	**96.09%**	1.86%	**95.62%**	3.24%	55.00%	84

The union passes every registered joint-objective gate relative to the frozen benchmark criteria; these gates do not require superiority over the privileged planner: 96.09% completion, 1.86% effective intervention, 95.62% warning recall, 3.24% warning FPR, and 84.50% fewer hazard-cost events than the naive controller (542 to 84). Its effective action-change recall is 55.00% (88/160): warned dangerous proposals do not count as interventions when the tangent command is already identical to the planner command. Executed intervention precision is 23.04% (88/382): 294 of 382 changed commands occur on oracle-labelled non-dangerous trajectories. That low precision is mechanistically consistent with the observed planner-relative hazard-cost increase, but the design does not identify causality and the paired interval includes zero. Its episode-bootstrap 95% intervals are 92.19%-99.22% for completion, 1.18%-2.55% for intervention, 81.58%-100.00% for warning recall, and 2.19%-4.31% for warning FPR. No final rerun or recovery path was used.

Relative to the privileged planner, the union changes completion by +1.5625 percentage points (paired 10,000-resample episode-bootstrap 95% CI [-1.5625, 4.6875]) and realized hazard cost by +14 steps (95% CI [-22.025, 54.000]). Neither interval excludes zero, so the observed two-task gain and 14-step increase are descriptive rather than statistically stable at this sample size.

Attribution remains essential. The privileged planner alone reduces hazard cost from 542 to 70 (87.08%). Adding the learned tangent branch increases completion from 121/128 to 123/128 but increases hazard-cost events from 70 to 84 (plus 14). All 382 effective union interventions are learned-only because the high-closeness rule never changes a command on this final distribution. The adapter warns on 153/160 dangerous controller trajectories, while 88/160 receive a different command; the remaining warned cases already propose the saturated tangent-compatible turn. The benchmark therefore supports a passing naive-baseline runtime objective and a completion/cost tradeoff over the planner, not learned collision-avoidance superiority over privileged planning.

8.5 Frozen risk-source attribution on fresh layouts

The v2 attribution opens seeds 8000-8127 once after committing every adapter, threshold, and simulator setting. The planner baseline records 122/128 completions, 116 hazard steps, and 7/128 hazardous episodes. Table 11 changes only the warning source; all rows retain the deterministic rule union and the same one-command tangent correction.

Table 11. Frozen risk-source attribution on seeds 8000-8127; only the warning source changes.

Risk source	Completion	Effective recall	Intervention	Precision	Hazard steps	Hazardous episodes	Mean steps
Full local + JEPA adapter	94.53%	47.53%	2.16%	24.27%	129	8/128	186.1
Local sensing, no JEPA	93.75%	46.62%	1.66%	31.71%	132	8/128	184.1
JEPA outputs only	96.09%	20.25%	1.35%	15.00%	104	7/128	184.9
Hazard closeness only	94.53%	51.08%	2.63%	22.83%	144	9/128	184.5

JEPA-only versus full was a prospectively specified member of a four-pipeline family, but not the uniquely designated primary contrast. The original 95% intervals are pointwise and unadjusted, so confirmatory interpretation would be inappropriate. A registered post-hoc sensitivity controls a 5% familywise error rate across the three non-full-versus-full pipeline contrasts within each endpoint separately; it does not adjust jointly across every endpoint or comparison in the report. For JEPA-only minus full, the Bonferroni-adjusted 98.33% interval is [-57, -3] for a -25-step hazard contrast and [-1.58, -0.01] for a -0.81-point intervention contrast. Both exclude zero. Hazard steps improve/worsen/remain unchanged in 6/0/122 paired episodes; intervention counts are lower/higher/unchanged in 26/16/86. Every one of 128 leave-one-seed-out 95% intervals still excludes zero for both quantities (hazard endpoints across omissions [-51, -3]; intervention endpoints [-1.52, -0.07]). These leave-one-out checks are sensitivity analyses, not independent replications. Its +1.56-point completion contrast has pointwise CI [0.00, 3.91], which shows no statistically resolved difference and is not a non-inferiority result. Against the unshielded planner, JEPA-only changes hazard cost by -12 steps with CI [-39, 4], so superiority to the planner is not established.

The original warning metrics are on-policy: each pipeline changes actions and therefore changes the states and proposals it later visits. Keeping the oracle logic fixed does not keep the evaluation population fixed. Table 12 reports both those on-policy values and a direct post-hoc detector comparison on the identical 23,815-proposal catalog visited by the full pipeline. The common catalog contains 263 dangerous and 23,552 safe proposals and reproduces the full arm exactly; it remains conditional on full-arm visitation and is not an independent sample.

Table 12. On-policy warning metrics and the post-hoc common-proposal detector comparison.

Risk source	Validation FPR	Final on-policy recall	Final on-policy FPR	Common recall	Common FPR
Full local + JEPA	0.40%	92.40%	3.35%	92.40%	3.35%
Local, no JEPA	0.37%	90.60%	2.53%	90.49%	2.41%
JEPA outputs only	0.43%	57.81%	2.66%	51.71%	2.90%
Hazard closeness	0.37%	80.58%	2.64%	79.85%	2.32%

Warning recall and realized outcomes rank differently: full has higher 20-step dangerous-trajectory warning recall, while JEPA-only intervenes less and records fewer realized hazard steps. The finding is exploratory pipeline-level evidence for a lower-intervention operating point under this fixed correction design. It does not isolate JEPA architecture, prove detector superiority on an independently sampled population, establish preserved completion, or establish safety superiority. The failed v1 attribution attempt and its excluded 7000-range partial catalog remain archived.

8.6 Retained-catalog Physical JEPA output-value ablation

The output-value ablation compares the same frozen model, adapter, operating threshold, planner, correction, and 128 seeds under two feature interventions. observed-frozen-jepa uses the three observed frozen JEPA outputs. development-mean-masked-jepa replaces only those values with the preregistered development means immediately before the unchanged adapter score. The observed arm completes 118/128 episodes with 173 hazard steps; the masked arm completes 119/128 with 183 hazard steps. The observed and masked catalogs contain 25,497 and 25,389 proposal rows respectively, so proposal-rate metrics are computed within arm while the uncertainty analysis pairs episodes by seed.

Table 13. Output-value ablation: observed minus development-mean-masked JEPA outputs, pointwise paired 95% intervals.

Observed minus development-mean masked	Estimate	Pointwise paired bootstrap 95% CI	Pointwise conclusion
Task completion, percentage points	-0.78125	[-2.34375, 0.00000]	Includes zero
Warning recall, percentage points	+3.02959	[-6.09578, 12.20268]	Includes zero
Warning FPR, percentage points	+1.66982	[0.95734, 2.47918]	Excludes zero
Effective intervention recall, percentage points	+0.50818	[-4.41887, 4.50930]	Includes zero
Intervention rate, percentage points	+0.96180	[0.58641, 1.38473]	Excludes zero
Intervention precision, percentage points	-15.25735	[-29.32521, -3.00212]	Excludes zero
Realized hazard-cost steps	-10	[-98, 59]	Includes zero

The seven displayed endpoints form the report-level multiplicity family. The post-hoc seven-endpoint Bonferroni-adjusted 99.29% intervals control 5% familywise error using the same 10,000 paired resamples and two-sided 0.003571 and 0.996429 quantiles:

Table 14. Output-value ablation with seven-endpoint Bonferroni-adjusted 99.29% intervals.

Observed minus development-mean masked	Bonferroni-adjusted 99.29% CI	Adjusted conclusion
Task completion	[-3.12500, 0.00000]	Includes zero
Warning recall	[-11.25330, 18.18255]	Includes zero
Warning FPR	**[0.78468, 2.81866]**	**Excludes zero**
Effective intervention recall	[-7.62912, 6.22118]	Includes zero
Intervention rate	**[0.47780, 1.57510]**	**Excludes zero**
Intervention precision	[-34.86661, 1.35705]	Includes zero
Realized hazard-cost steps	[-134, 81]	Includes zero

Observed JEPA output values therefore produce more false-positive warnings and more effective action changes in this fixed closed-loop software pipeline. Warning FPR and intervention rate remain separated from zero after adjustment, and neither is evidence of benefit. The lower precision appears only in the pointwise analysis; its adjusted interval includes zero. Completion, warning recall, effective recall, and realized hazard cost remain unresolved. The -10 observed hazard-step estimate is especially not a superiority result because its interval is wide and includes zero. As a stricter family-definition audit, adjustment across all 11 registered paired outputs gives the same survivor set: warning FPR [0.73638, 2.89725] and intervention rate [0.45556, 1.59936].

These tables report a retrospectively recovered analysis of complete catalogs produced by a prospectively specified and executed experiment that failed its final protected-file check. The failed v1 result remains ineligible; recovery v2 did not rerun the simulator; verification v3 recomputes the catalogs, scores, aggregates, paired effects, seeds, and feature intervention and passes all 51 checks. The multiplicity sensitivity is post-hoc and changes no registered endpoint, seed, or bootstrap

replicate. The evidence supports only the direct effect of observed-versus-masked output values in this fixed local pipeline. It does not provide a successful prospective result, architecture-wide causality, independent assessment, HIL, physical safety, deployment safety, or promotion eligibility.

8.7 Registered incident-motivated prediction and warning improvement (v2)

Question and evidence class. This additional study changes development data, training objectives, model families, and calibration together. It tests a better prediction-and-warning pipeline, not an isolated JEPA innovation or a controller. Eight primary-source narratives motivate synthetic pressure, degradation, recovery, and stale-observation patterns. They supply neither numeric transition coefficients nor ground-truth labels.

Table 15. Incident narratives used in the v2 study and the abstract perturbation each motivates.

Narrative sources	Role	Representation and boundary
Google Rapid Reset; Cloudflare configuration outage [20, 21]	OS development	Repeated resource work and coupled growth; no HTTP replay or configuration-parser model
Siemens advisory; NASA Ingenuity investigation [22, 23]	Physical development	Degraded communication and stale observations; no vendor exploit or aircraft model
Cloudflare MadeYouReset; AWS DynamoDB outage [24, 25]	OS final source labels	Altered pressure/recovery timing using shared equations; not a novel-mechanism holdout
Rockwell advisory; NTSB San Antonio investigation [26, 27]	Physical final source labels	Stronger late degradation and stale clearance; not road-driving or vendor-security validation

The advisories do not establish exploitation, and the outages are not classified as attacks. NASA's likely navigation explanation retains investigative uncertainty. Source identities are assigned to generated episodes; they do not cause distinct learned transition laws. With two source labels assigned by episode index modulo two and four pressure regimes assigned modulo four, source-label strata are confounded with pressure regime; they cannot estimate source-specific transfer. Final OS recovery begins at step nine with a reversed half-strength drift rather than the development stop at step eleven; physical degradation increases by 50% after step seven. Thus the fresh test changes timing and severity within a researcher-designed generator, not the underlying independent source of dynamics. No incident text is treated as measured state-action telemetry.

Development and selection. Original development partitions are augmented with 1,200 new 14-step episodes per domain (seed 281001); 256 additional validation episodes use seed 281002. There are 30,512 OS and 204,000 physical training windows, and 4,028 and 25,280 complete validation windows. Inputs contain three causal state/action observations and the proposed action sequence, never future states, exogenous drift, or hazard labels. Short training tails are masked. Neural candidates minimize normalized smooth-L1 rollout error at all five future steps without teacher forcing after the first prediction; the JEPA control also retains a 0.01-weighted latent loss. AdamW uses learning rate 0.001, batch 256, gradient clipping, and deterministic cosine decay. Thirteen completed neural trials, all checkpoint traces, and ridge regularization trials are retained. Selection minimizes mean validation H=1/3/5 episode-macro MAE and averages every completed seed within a candidate configuration; it never chooses a lucky individual seed.

The selected OS model is per-action ridge regression with 621,888 fitted coefficients and regularization 0.001. Its weights are learned from one-step transitions, not copied from simulator equations. The physical model averages seeds 17, 43, and 101 of an action-residual network, each with 73,840 parameters and 12,000 updates: a shared history encoder feeds learned action-specific delta heads. The strongest retained JEPA and MLP controls use the same development data and multi-step objective, but have different parameter counts, completed seeds, and budgets. The study is not architecture-only, parameter-matched, or FLOP-matched. Validation traces describe training variability; test intervals condition on the fixed selected ensemble rather than resampling training runs.

Registration and failure boundary. Protocol v1 and its selection were committed before final access. Its first attempt generated OS arrays in memory, then rejected duplicate historical step identifiers before final model inference or saved catalogs. That failed attempt remains ineligible. Amendment v2, committed as 3086e52, excludes every ambiguous duplicate-step historical episode before scoring and uses fresh seed 281014. It reuses identical selected models, calibration, thresholds, effect gates, and bootstrap settings; selection reuse was committed as 645fd8f before the v2 opening. Exactly one OS episode is excluded; it has only two rows and was already ineligible for the five-step population. No physical episode is excluded. The successful v2 final is not a rerun of v1 evidence.

Frozen comparison. The new catalog has 1,024 episodes and 10,240 common five-step-eligible windows per domain. Primary prediction gates require at least 20% lower normalized MAE at H=3 and H=5 in both domains, with 10,000 paired episode-bootstrap 98.75% intervals excluding zero (Bonferroni across four endpoints). The primary comparators are the original paper-selected OS GRU and physical JEPA ensembles; all nine required legacy checkpoints were reproduced byte-exactly and archived. Historical final families, already opened by the earlier paper, are regression checks only. Their prespecified point-estimate guard allows at most 5% error increase against the best available paper ensemble on the common H=5-eligible episode population; it is not statistical non-inferiority.

Table 16. Incident-dynamics v2 primary prediction endpoints on the fresh v2 catalog.

Fresh-catalog endpoint	Earlier paper ensemble	Selected model	Relative reduction	Selected minus paper, 98.75% CI
OS H=3	0.046242	0.000273	99.41%	[-0.046133, -0.045806]
OS H=5	0.047111	0.000511	98.92%	[-0.046779, -0.046425]
Physical H=3	0.015085	0.006667	55.80%	[-0.008621, -0.008206]
Physical H=5	0.022689	0.009369	58.71%	[-0.013665, -0.012970]

All four primary gates pass. The large OS reduction is against an earlier ensemble shifted onto a new sparse-state catalog; many state coordinates are inactive, so the headline is not sufficient by itself. At H=5 on the three OS warning dimensions, MAE falls from 0.063280 to 0.007786 (87.70%). The following complete control table gives the more informative comparison with retraining and simple extrapolation. These control contrasts are descriptive, not additional multiplicity-adjusted discoveries.

Table 17. Incident-dynamics v2 control comparison at H=5 on the fresh v2 catalog.

Method	OS H=5 normalized MAE	Physical H=5 normalized MAE
Original paper GRU	0.047111	Not registered
Original paper JEPA	0.048138	0.022689
State persistence	0.001145	0.056278
Causal constant velocity	0.001568	0.164833
Same-data retrained JEPA	0.001251	0.011077
Same-data retrained MLP	0.001201	0.012153
Per-action ridge	**0.000511**	0.019034
Selected physical residual ensemble	Not selected	**0.009369**

At H=5, the strongest non-selected controls in this table are OS persistence and physical retrained JEPA. The selected models reduce error by 55.35% and 15.42% against these controls, respectively. Those comparisons were not the registered primary family: in particular, the physical comparison does not meet the 20% primary effect size, and no adjusted superiority claim against that control is made.

On the historical common-episode population, all six regression guards pass. OS H=1/3/5 MAE decreases by 88.20%/33.59%/41.42% against the best available paper ensemble; physical decreases by

63.64%/71.08%/75.19%. These are recomputed comparisons on 238 OS and 2,560 physical episodes, not replacements for the earlier differently composed table and not fresh confirmation.

Warning results and cost. Each pipeline receives its own calibration on 512 separate episodes (seed 281003): an increasing logistic map for probabilities and the smallest score threshold yielding at most 1% empirical calibration FPR. Test labels mean any represented constraint breach in the next five states: OS resource coordinates 0/1/3 at least 0.95; physical boundary magnitude at least 1, battery or link at most 0.08, or moving clearance at most 0.16. The secondary gate requires at least 20 percentage points higher recall, a positive paired 97.5% interval across the two recall endpoints, and observed final FPR at most 2%. It does not require equal final FPR or guarantee the 1% calibration target under shift.

Table 18. Incident-dynamics v2 warning results with independent calibration.

Domain / pipeline	TP / FP / TN / FN	Recall	FPR	Brier
OS paper GRU, newly calibrated	191 / 59 / 8731 / 1259	13.17%	0.67%	0.075390
OS selected ridge	1365 / 147 / 8643 / 85	**94.14%**	1.67%	**0.018346**
Physical paper JEPA, newly calibrated	1460 / 71 / 6481 / 2228	39.59%	1.08%	0.104650
Physical selected ensemble	3151 / 93 / 6459 / 537	**85.44%**	1.42%	**0.037642**

Recall gains are 80.97 points [77.08, 84.68] and 45.85 points [42.12, 49.56] at 97.5% confidence; both warning gates pass. This is not cost-free: OS FPR rises by 1.00 point [0.45, 1.59], while the physical increase of 0.34 point [-0.08, 0.79] is unresolved. These FPR intervals are descriptive and not included in the registered recall multiplicity family. On the currently observed-safe subset, selected recall is 87.48% for OS and 77.16% for physical, so the result is not solely detection of already-visible constraint breaches. Ten-bin ECE is 0.007886 and 0.009405; the associated prevalences are 14.16% and 36.02%, and complete reliability bins and source-label strata remain in the result JSON.

Interpretation and audit. Incident-motivated training and model selection substantially improve prediction and thresholded warnings within this simulator family. The experiment does not identify the separate causal contribution of additional data, multi-step loss, architecture, ensemble size, or calibration. It does not evaluate applied interventions, realized hazard reduction, Safety-Gymnasium controller improvement, or live incidents. The earlier zero-intervention and unfavorable output-ablation results therefore stand unchanged. All 48 retained-catalog checks pass in the working tree and a clean source archive; the portable wrapper permits only the already registered LF/CRLF equivalence for JSON metadata. Checkpoints, NPZ catalogs, and the archived earlier PDF remain byte-exact. No new model replaces a deployed binary and no actuator authority is enabled.

8.8 Incident-dynamics v3: broader mechanisms, boosted residual models, and an evolving-world curriculum

Question and evidence class. Section 8.7 improved prediction and warnings against the earlier paper ensembles. This study asks whether the models selected there can themselves be improved, on two registered populations: (A) a fresh seed from the unchanged v2 final generator, which is the population behind Tables 16-18, and (B) mechanism families held out entirely from development. The comparator is the v2-selected pipeline: per-action ridge for FerrumOS and the three-seed action-residual ensemble for Physical. The evidence class is unchanged: a prospective, locally designed software benchmark with no authority change.

Sources and mechanisms. Twenty-four newly verified primary sources motivate ten abstract mechanism families per domain (Table 19). OS sources include official root-cause analyses and summaries (CrowdStrike, AWS, Cloudflare, GitLab, Meta, Instapaper), a regulator order (SEC, Knight Capital), a peer-reviewed measurement study (Mirai), and vulnerability guidance (CISA Log4Shell; the xz-utils disclosure). Physical sources include CISA alerts, NTSB investigations, NASA and ESA mishap

reports, an EASA airworthiness directive on GNSS interference, the Stuxnet dossier, MITRE ATT&CK, TRITON threat research, and the Jeep Cherokee remote-compromise paper. A narrative fixes only a qualitative signature such as a monotone leak, a retry oscillation, a replayed sensor, or an actuator gain error. Every coefficient, range, label, and equation is researcher-authored, and no source supplies telemetry. Titles, publishers, dates, URLs, and caveats are in incident_dynamics_sources_v3.json.

Table 19. Incident-motivated mechanism families, their partition, and motivating sources.

Domain	Mechanism family	Partition	Motivating sources	Abstract perturbation
OS	constant pressure	development	v1/v2 sources; GitLab 2017	per-episode constant drift on process/heap/disk (v2 pressure regimes)
OS	accelerating leak	development	AWS 2012	heap (and optionally disk) growth whose per-step increment rises linearly
OS	exponential amplification	held out	SEC Knight Capital; Mirai (USENIX)	process count grows geometrically (retry or fork amplification) from a small seed
OS	sudden step	development	Cloudflare 2019	one-time jump in heap or process count at a random step (faulty update or config push)
OS	retry oscillation	development	AWS 2021	periodic process/heap load (retry storm or thundering herd)
OS	coupled cascade	development	AWS S3 2017	process growth induces lagged heap growth that later drives disk writes
OS	crash restart	held out	CrowdStrike RCA 2024	rising heap until a crash drops processes and heap, then gradual restart
OS	log storm	development	Instapaper 2017	file count and disk usage grow together (log or temp-file storm)
OS	frozen telemetry	held out	Meta 2021	monitoring stalls: one resource reading repeats for several steps
OS	regime shift	development	CISA Log4Shell; xz-utils disclosure	drift changes sign or magnitude once at a random step (mitigation or escalation)
Physical	constant pressure	development	v1/v2 sources	per-episode link and battery drift (v2 pressure regimes)
Physical	accelerating drain	development	NTSB JAL 787	battery drain increment rises over time (aging or thermal stress)
Physical	link jamming	development	EASA GNSS AD	intermittent bursts of severe link degradation with partial recovery
Physical	actuator gain fault	held out	Jeep Cherokee paper; NASA MCO; ESA Ariane 501	commanded displacement is scaled by an unknown factor (degraded motor or unit error)
Physical	approaching obstacle	held out	NTSB Uber ATG	clearance shrinks independently of own motion (moving obstacle)
Physical	replayed sensors	held out	Stuxnet dossier	position and clearance readings replay an earlier segment (replay attack)
Physical	position bias	development	NTSB 737 MAX	slowly growing additive bias on observed position (sensor fault or spoofing)
Physical	vibration degradation	development	generic wear (no incident claimed)	rising vibration and falling health toward a fault
Physical	regime shift	development	CISA Oldsmar; MITRE Maroochy; TRITON	link or battery drift changes magnitude once at a random step
Physical	stale observation	development	CISA Ukraine 2015	periodic stale clearance and position readings

Novel-mechanism holdout. Three families per domain are held out, and their motivating sources are disjoint from development sources. The generator refuses to emit a held-out family outside the registered final partition. The development plan discloses two pre-plan probes: a noise-only oracle on the public generators, and a sanity check that generated every family (including held-out ones) and

reported only counts, prevalence, and finiteness. No model was scored on held-out or final-generator samples before the single final opening.

Development and selection. Development uses the unchanged v2 partitions, whose history-3 windows are verified byte-identical to v2, plus 400 training and 64 validation episodes per development family. Nineteen v3 configurations were retained (Appendix D, Table D3): autoregressive residual networks, network and gradient-boosted [28] residual corrections on top of the frozen v2-selected model, an effect-drift network with in-context drift identification, and a learned warning-scale add-on. Longer-history ridge and ridge-plus-network hybrids were explored in development probes; the hybrid and uncertainty-aware network trials were lost to GPU faults or stopped when the boosted families dominated, and are logged rather than retained. Selection minimizes the mean, over legacy, v2-incident, and development-mechanism validation blocks, of each block score divided by the v2-selected block score, so v2-selected scores exactly 1.0 and no block dominates by error scale. This metric replaced an earlier unbalanced one before any selection; the amendment and the three trials scored under the old metric are recorded. A configuration needs three seeds. The selected FerrumOS model applies direct per-horizon LightGBM corrections to the frozen ridge rollout (score 0.690); the selected Physical model applies one-step LightGBM corrections to the frozen v2 ensemble (score 0.649). Neural corrections lost precision on the v2-incident block (scores 0.94-1.37 for FerrumOS); a one-step boosted FerrumOS model was 46-67% better than ridge on five-step windows that end before an unannounced drift change but 52-70% worse on windows that cross it, which the direct per-horizon form removed by learning how far to extrapolate drift, much like a damped trend [29]. The learned warning scale lowered Physical development recall at matched FPR, so the registered rule chose k=0 for both domains.

Evolving-world curriculum. Dream-RSI optimizes an agent's exploration policy by replaying stored discovery trees [18]. Our adaptation keeps its loop structure but applies it to training worlds. Round 0 fits on exactly the uniform development data. Each later round probes mutated development worlds with the current model, adds the three it predicts worst relative to persistence, fits three candidate continuations under different pool weightings, and keeps the candidate with the best replay score over the stored validation samples of every world, with the unchanged current model always eligible. The registered ablation compares this arm with the uniform counterpart of the same FerrumOS family and seeds. The arms are not compute-matched, and no physical arm was run.

Registered outcome. All eight primary endpoints improve on the v2-selected models with Bonferroni-adjusted 99.375% intervals excluding zero (Table 20), but only Physical population A meets the registered 20% effect size. The Physical historical H=5 guard also fails: one-step corrections compound across five steps on the earlier paper families, raising H=5 error by 7.9% against v2-selected while H=1 and H=3 fall by 63.4% and 24.2%. The primary success rule therefore fails, and the v3 result is retained as a failure.

Table 20. Incident-dynamics v3 primary endpoints against the v2-selected models (lower is better).

Population	Domain	Horizon	v2-selected MAE	v3-selected MAE	Reduction	v3 minus v2, 99.375% CI	Gate
A	OS	H=3	0.000269	0.000225	16.3%	[-0.000047, -0.000041]	fail
A	OS	H=5	0.000499	0.000468	6.3%	[-0.000036, -0.000026]	fail
A	Physical	H=3	0.006600	0.004368	33.8%	[-0.002509, -0.001981]	pass
A	Physical	H=5	0.009249	0.006618	28.4%	[-0.003043, -0.002238]	pass
B	OS	H=3	0.000887	0.000742	16.4%	[-0.000156, -0.000134]	fail
B	OS	H=5	0.001667	0.001361	18.4%	[-0.000329, -0.000283]	fail
B	Physical	H=3	0.008811	0.007224	18.0%	[-0.001851, -0.001343]	fail
B	Physical	H=5	0.012080	0.010273	15.0%	[-0.002193, -0.001438]	fail

FerrumOS population A shows why the five-step gain is small. Every A episode reverses its drift at step nine, so half of the five-step windows cross a change that no causal model can anticipate; v3 is 16.3% better at H=3 but only 6.3% better at H=5. Against simple controls the gains are larger (Table 21): at H=5 v3 is 58.3% below persistence on OS population A and 88.2% below persistence on Physical population A.

Table 21. H=5 normalized MAE on the v3 final populations for every registered pipeline.

Method	OS A	OS B	Physical A	Physical B
State persistence	0.001123	0.002903	0.056037	0.057246
Causal constant velocity	0.001550	0.002452	0.165764	0.168603
Paper GRU ensemble	0.047127	0.046822	not run	not run
Paper JEPA ensemble	0.048046	0.047854	0.022815	0.020999
v2 per-action ridge	0.000499	0.001667	0.019018	0.019401
v2-selected (Section 8.7)	0.000499	0.001667	0.009249	0.012080
v3 uniform curriculum	0.000468	0.001361	not run	not run
v3 evolving-world curriculum	0.000474	0.001382	not run	not run
v3-selected	0.000468	0.001361	0.006618	0.010273

Warnings. Recall rises in three of four cells, most strongly for FerrumOS (93.2% to 98.6% on A and 60.0% to 76.7% on B), but observed FerrumOS FPR (2.02% and 2.12%) exceeds the registered 2% cap, so those cells fail. Physical A passes with 22.0% fewer missed hazards at 0.95% FPR; Physical B does not (3.2%). The secondary warning objective therefore fails (Table 22).

Table 22. Incident-dynamics v3 warnings; every pipeline is recalibrated on the same calibration episodes.

Population	Domain	Pipeline	TP / FP / TN / FN	Recall	FPR	Brier	Miss reduction	Gate
A	OS	v2-selected, recalibrated	1183 / 132 / 8839 / 86	93.2%	1.47%	0.0227	-	-
A	OS	v3-selected	1251 / 181 / 8790 / 18	98.6%	2.02%	0.0194	79.1%	fail
A	Physical	v2-selected, recalibrated	2860 / 63 / 6776 / 541	84.1%	0.92%	0.0364	-	-
A	Physical	v3-selected	2979 / 65 / 6774 / 422	87.6%	0.95%	0.0301	22.0%	pass
B	OS	v2-selected, recalibrated	674 / 106 / 9011 / 449	60.0%	1.16%	0.0413	-	-
B	OS	v3-selected	861 / 193 / 8924 / 262	76.7%	2.12%	0.0412	41.6%	fail
B	Physical	v2-selected, recalibrated	3147 / 95 / 6102 / 896	77.8%	1.53%	0.0718	-	-
B	Physical	v3-selected	3176 / 86 / 6111 / 867	78.6%	1.39%	0.0699	3.2%	fail

RSI ablation. The evolving-world arm improved its own replay objective in every round, but on the registered populations it was 1.3% (A) and 1.6% (B) worse than the uniform counterpart at H=5, with paired intervals excluding zero in that direction (Table 23). Under the registered claim rule, v3 provides no evidence of an RSI gain: targeting the hardest synthetic worlds did not transfer to held-out mechanisms.

Table 23. FerrumOS evolving-world (Dream-RSI-inspired) arm versus its uniform counterpart.

Population	Horizon	Uniform MAE	Evolving-world MAE	Reduction	Paired 95% CI (RSI minus uniform)
A	H=3	0.000225	0.000225	0.1%	[-0.000001, 0.000000]
A	H=5	0.000468	0.000474	-1.3%	[0.000005, 0.000007]
B	H=3	0.000742	0.000739	0.4%	[-0.000007, 0.000001]
B	H=5	0.001361	0.001382	-1.6%	[0.000013, 0.000029]

Audit. The verifier regenerates every registered catalog from its seed, reloads the compact archived checkpoints, recomputes inference and statistics, and passes all 48 checks. Boosted checkpoints are archived as prediction-only LightGBM text compressed with xz after selection asserted bit-identical predictions. Seven laptop-GPU "illegal instruction" faults (three in the first unqueued concurrent batch), two RAM exhaustions, one RSI run killed with its queue runner and rerun, and the mid-development stop of the neural queues are logged in infrastructure_failures.jsonl and the development ledger. No checkpoint replaces a deployed binary, and no actuator authority is enabled.

8.9 Incident-dynamics v4: a registered successor after the failed v3 rule

Design. v3 is retained as a failure. Its Physical guard failure pointed to a specific, fixable weakness: one-step corrections compound across five steps. v4 therefore adds a Physical direct multi-horizon boosted family (300 trees per horizon and dimension) and a registered committee, carries the FerrumOS models forward unchanged, keeps every v3 gate, and opens fresh populations A (seed 391024) and B (seed 391025). v4 is a registered successor informed by v3 results, not an independent confirmation of v3. No v3 final catalog entered v4 development, selection, or calibration. On development data the direct Physical ensemble scored 0.579, ahead of the committee (0.600) and the v3 one-step model (0.649), and was selected.

Outcome. Physical meets every registered gate: H=3 and H=5 error fall 38.8% and 38.0% on population A and 23.8% and 24.1% on population B, and the historical guard turns from a 7.9% H=5 regression into a 55.3% improvement (Tables 24 and 25). FerrumOS population B also passes (20.0% and 20.8%). FerrumOS population A does not: 15.6% and 5.7%, with intervals excluding zero. The primary rule therefore fails on that one population, for the reason identified in Section 8.8: every A episode reverses its drift at step nine, after the last observation available to any five-step window that crosses it. Raising that number would require training on the known reversal of the test generator, which this study does not do.

Table 24. Incident-dynamics v4 primary endpoints against the v2-selected models.

Population	Domain	Horizon	v2-selected MAE	v4-selected MAE	Reduction	v4 minus v2, 99.375% CI	Gate
A	OS	H=3	0.000264	0.000223	15.6%	[-0.000044, -0.000039]	fail
A	OS	H=5	0.000493	0.000465	5.7%	[-0.000033, -0.000023]	fail
A	Physical	H=3	0.006697	0.004101	38.8%	[-0.002849, -0.002360]	pass
A	Physical	H=5	0.009565	0.005935	38.0%	[-0.004016, -0.003295]	pass
B	OS	H=3	0.000763	0.000610	20.0%	[-0.000162, -0.000143]	pass
B	OS	H=5	0.001542	0.001221	20.8%	[-0.000342, -0.000300]	pass
B	Physical	H=3	0.009048	0.006899	23.8%	[-0.002443, -0.001896]	pass
B	Physical	H=5	0.012484	0.009471	24.1%	[-0.003443, -0.002624]	pass

Table 25. Physical v4 direct model against the v2-selected and v3-selected models, and the historical guard.

Evidence	Horizon	v2-selected	v3-selected (one-step)	v4-selected (direct)	v4 vs v2	v4 vs v3
Population A	H=3	0.006697	0.004444	0.004101	38.8%	7.7%
Population A	H=5	0.009565	0.006791	0.005935	38.0%	12.6%
Population B	H=3	0.009048	0.007245	0.006899	23.8%	4.8%
Population B	H=5	0.012484	0.010393	0.009471	24.1%	8.9%
Historical guard	H=1	0.000900	-	0.000323	64.1%	-
Historical guard	H=3	0.001932	-	0.000803	58.4%	-
Historical guard	H=5	0.002597	-	0.001160	55.3%	-

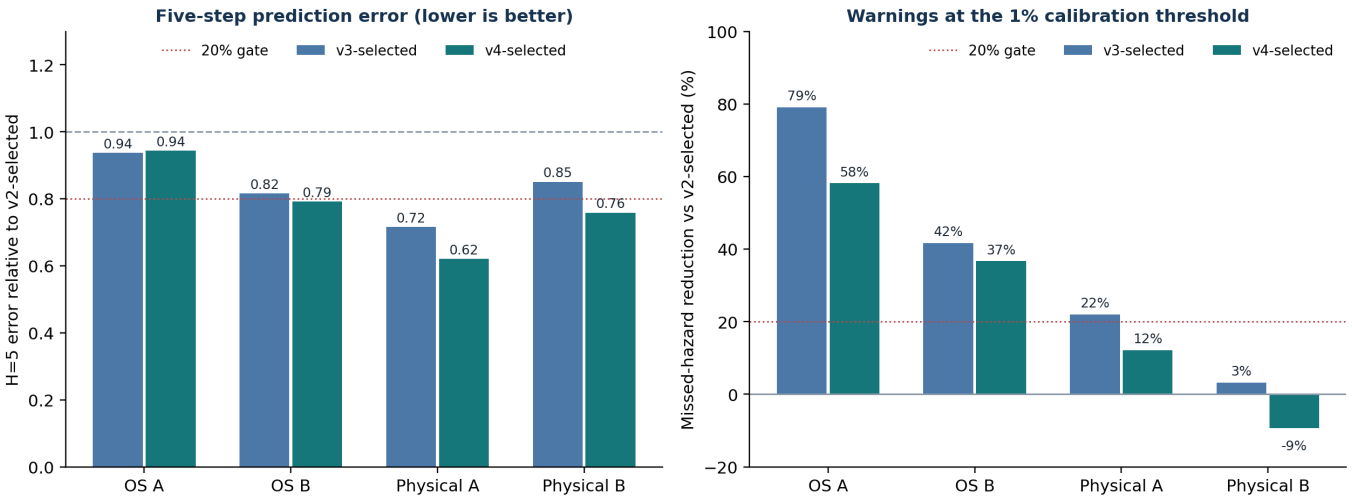


Figure 5. Incident-dynamics v3 and v4 against the v2-selected models on each study's own fresh populations. Left: H=5 error ratio (0.8 is the registered 20% gate). Right: reduction in missed delayed-constraint hazards at the 1% calibration threshold.

Warnings do not follow the prediction gains. FerrumOS B passes the warning gate (36.7% fewer misses at 1.91% FPR). FerrumOS A recall rises from 92.0% to 96.7%, but observed FPR is 2.42%, above the 2% cap. Physical warnings barely move: 12.1% fewer misses on A and 9.3% more on B (Table 26). Lower mean error does not by itself sharpen a thresholded constraint margin on noisy dimensions, and the learned warning scale did not help on development data. The secondary warning objective fails.

Table 26. Incident-dynamics v4 warnings; every pipeline is recalibrated on the same calibration episodes.

Population	Domain	Pipeline	TP / FP / TN / FN	Recall	FPR	Brier	Miss reduction	Gate
A	OS	v2-selected, recalibrated	1490 / 161 / 8460 / 129	92.0%	1.87%	0.0279	-	-
A	OS	v4-selected	1565 / 209 / 8412 / 54	96.7%	2.42%	0.0231	58.1%	fail
A	Physical	v2-selected, recalibrated	2616 / 110 / 6946 / 568	82.2%	1.56%	0.0412	-	-
A	Physical	v4-selected	2685 / 114 / 6942 / 499	84.3%	1.62%	0.0350	12.1%	fail
B	OS	v2-selected, recalibrated	695 / 98 / 8992 / 455	60.4%	1.08%	0.0434	-	-
B	OS	v4-selected	862 / 174 / 8916 / 288	75.0%	1.91%	0.0422	36.7%	pass
B	Physical	v2-selected, recalibrated	3130 / 80 / 6152 / 878	78.1%	1.28%	0.0708	-	-
B	Physical	v4-selected	3048 / 75 / 6157 / 960	76.0%	1.20%	0.0712	-9.3%	fail

Audit. The v4 runner is a thin wrapper around the frozen v3 runner, so the v3 bindings stay reproducible; reused FerrumOS checkpoints point to the v3 compact archives after a bit-identity check. The verifier regenerates every v4 catalog, recomputes inference and statistics, and passes all 46 checks.

9. Cross-domain synthesis

9.1 Prediction quality is domain-specific

The registered table rejects a universal architecture story. Physical dynamics favor the tested three-member JEPA ensemble at all horizons. FerrumOS favors the tested three-member GRU ensemble at H=1 and H=3, with JEPA leading at H=5; however, the common-H5-eligible episode sensitivity favors JEPA at every horizon. The apparent FerrumOS reversal is therefore composition-sensitive and cannot be isolated as a horizon effect. This makes architecture selection an empirical, estimand-specific decision rather than a model-family claim. The incident-dynamics studies add a second pattern: in these compact, structured state spaces, boosted corrections on a frozen simple model beat every neural family tried, and per-horizon models beat one-step rollouts at five steps. That is a finding about these simulators, not a general ranking of model families.

9.2 Causal response is not calibrated authority

Both originally selected models react to paired interventions in the intended direction. Neither generates an operational intervention at the conservative frozen threshold on the original shifted final catalogs. These findings are compatible: a model can encode direction and approximate magnitude while its calibrated score lacks separation at a chosen false-positive constraint. Section 8.7 improves warnings under a different trained pipeline, population, and calibration procedure; it does not retroactively repair that original operational result or demonstrate beneficial interventions. Reporting only rollout or directional accuracy would conceal the distinction.

9.3 Deterministic authority can fail differently

The original 512-case catalogs show no interventions from either rules or learning, while the 3D stress intervenes on every case. The prospective Safety-Gymnasium result separates a useful privileged planner-only controller from an active union that passes the naive-baseline objective but trades higher completion for higher hazard cost relative to the planner. The retained-catalog output intervention then shows that observed JEPA values increase warning FPR and intervention rate under both pointwise and multiplicity-adjusted intervals. Precision decreases only pointwise; its seven-endpoint adjusted interval includes zero, as do completion and realized hazard cost. These contrasts show why controller quality cannot be credited to a shield and why warning recall or intervention count cannot substitute for marginal executed outcomes. Deterministic authority must be evaluated with task completion, intervention, proposal recall, false positives, precision, realized cost, and controller divergence rather than a collision count alone.

9.4 Evidence ladders prevent category errors

QEMU timing is not production throughput. Researcher-operated sessions are not independent user evidence. Recorded sensors are not live HIL. PyBullet contact is not physical collision evidence. An externally maintained simulator task executed by the author is not an independent replication, and a planner with direct simulator geometry is not a sensor-only robot controller. A publisher-heldout robotics file is not compatible merely because it contains robot telemetry. Labelling those boundaries does not weaken the contribution; it makes the evidence composable and reproducible.

9.5 Novelty statement

This work does not claim the first JEPA world model, safety filter, calibrated predictor, or runtime-assurance architecture. Its central novelty is an executable evaluation method with named estimands and named failure modes: prediction error, counterfactual response, calibration, warnings, effective interventions, realized outcomes, and authority are registered and reviewed separately. Applying that method in two domains exposes a composition-sensitive registered ranking, a thresholded latent-hazard zero, an all-stop policy, a planner-dominated result, and a risk-source attribution that warning recall alone would conceal. The cross-domain claim is methodological replication across distinct systems, not learned representation or policy transfer.

10. Threats to validity and limitations

1. The temporal catalogs and PyBullet stress use locally designed deterministic labels. Safety-Gymnasium improves task provenance, but adapter design, execution, and assessment remain local; no independent replication is claimed.
2. The architecture study matches data, seeds, parameters, and updates, not FLOPs or training wall time. Its episode bootstrap conditions on fixed trained checkpoints and omits retraining variability. Registered horizon populations differ; the common-episode post-hoc sensitivity changes the FerrumOS ranking, so the registered reversal cannot be isolated as a horizon effect.
3. The attribution varies complete risk-source pipelines, not architecture alone. JEPA-only retains a historically trained v5 artifact, so its lower hazard count cannot be attributed purely to a JEPA objective. JEPA-only versus full was one prospectively specified family member, not a unique primary contrast. The Bonferroni sensitivity is familywise across three non-full-versus-full pipeline contrasts within each endpoint, not across every endpoint or comparison; the leave-one-out checks are sensitivity analyses, not independent replications.
4. The original temporal calibration is weak under shift. Its constant-prevalence Brier baseline is 0.25, ECE is bin-dependent, and risk-coverage ordering is non-monotonic. Zero observed false positives are not a population guarantee. Section 8.7 uses a different population and calibration protocol, with its own prevalence, Brier score, and observed FPR cost.
5. The 512-case families place danger beyond present-state rules by construction. Their rules-only zero is catalog coverage, not general rule quality; the [0,0] bootstrap reflects an all-zero observed vector.
6. FerrumOS runtime evidence uses QEMU/WHPX, one serial daemon, four local clients, and 24 requests from one researcher. No production, multi-host, independent-user, or long-duration evidence exists.
7. Timing uses different clocks and workloads. No synchronized model/queue/transport/polling decomposition or calibrated natural-use cycle conversion is available.
8. The HAI replay projects eight recorded channels into a 16-state representation using frozen training statistics. Labels and proxy actions are not equivalent to the model ontology; it is replay, not live HIL.
9. Anchor-Lab telemetry is externally authored but semantically incompatible with direct v5 scoring. It supports future embodiment-specific work, not present-model transfer.
10. The PyBullet environment is locally designed and simple. Its 100% intervention and 0% completion remain a negative stress result, not useful collision avoidance.
11. The strongest Safety-Gymnasium controller uses privileged simulator geometry. Planner effects must not be credited to the shield, and the fixed rule's inactivity is distribution-specific.

12. Warning recall uses a 20-step nominal-controller oracle. It is dangerous-trajectory warning recall, not a generic system-safety metric; effective action recall and realized cost are separate. Original arm values are on-policy and therefore use different visited proposals. The common-proposal check is conditional on the full-arm catalog, not an independently sampled detector benchmark.

13. The output-value ablation's design, estimands, fresh seed range, and bootstrap seeds were registered before either catalog existed, and both catalogs completed during the prospective v1 execution. Its result nevertheless failed eligibility because an unrelated concurrent rebuild changed a protected report PDF before the final integrity check. Recovery v2 is a subsequently registered retrospective reportable analysis of those retained catalogs, not a successful prospective result or rerun. The seven-endpoint and stricter 11-output multiplicity sensitivities are post-hoc. All paired estimates remain conditional on the fixed seed set, planner, correction, threshold, and local simulator pipeline; the intervention does not isolate architecture or establish benefit.

14. Source tests cover enumerated protocol, daemon, and physical-runtime paths. Deterministic predicates are engineering rules, not formally verified invariants, and empirical tests do not prove universal non-bypass.

15. There is no live actuator timing, physical contact, hardware emergency stop, human-contact dynamics, or independent execution. All research artifacts remain ineligible for promotion and protected deployed artifacts are unchanged.

16. The incident-motivated study uses researcher-authored equations, perturbation ranges, labels, and analysis. Narrative-source holdout is not independent dynamics or telemetry holdout; source labels do not instantiate separate transition laws. The extreme OS relative gain partly reflects the old model's mismatch to a sparse synthetic population, making persistence, same-data controls, and warning-dimension errors essential context.

17. New pipeline budgets and seed counts differ, so additional data, objective, architecture, ensembling, and calibration effects are not separately identified. Bootstrap intervals condition on selected checkpoints. Historical checks are previously exposed regression populations, not new confirmation. The two-domain warning gate constrains observed FPR, not a population upper confidence bound; the significant OS FPR increase remains a cost.

18. Incident-dynamics v1 failed on historical input schema before final model scoring. V2 is a prospectively registered fresh-seed amendment with unchanged selection and thresholds, not a successful v1 result. Its prediction and warning success is not an intervention-outcome, exploit-detection, physical-safety, or Dream-RSI result.

19. The v3 mechanism library, strength ranges, constraint labels, and analysis are researcher-authored. The novel-mechanism holdout is within one simulator family: held-out families share the state representation, the action ontology, and the labels, and their motivating narratives supply no telemetry.

20. v3 corrections are learned on top of the frozen v2-selected models. Gains belong to the whole pipeline (mechanism data, eight-step history, incident weighting, boosted direct or one-step corrections, three-seed averaging); no architecture-only or compute-matched claim is made, and the boosted models are larger than the corrected base.

21. Population A is not a timing holdout for v3 models because the development curriculum includes regime-change families. Its five-step gain is bounded by an unannounced reversal that every causal model shares. The v3 primary rule failed on six of eight effect sizes and one historical guard, and that failure is retained rather than reclassified.

22. The evolving-world arm adapts only Dream-RSI's loop structure to training worlds; it is FerrumOS-only, not compute-matched, and its registered comparison is negative. It is not a test of Dream-RSI's exploration-policy optimization.

23. v4 is a registered successor designed after the v3 result. Its populations are fresh and no v3 final catalog entered its development, but its added family was chosen because of v3's Physical guard failure, so v4 is not an independent confirmation of v3. FerrumOS v4 re-measures the unchanged v3 models on a new sample.

11. Reproducibility and artifact integrity

Every evidence object names its scope and digest. Hashes prove byte identity only; they do not validate labels, scientific assumptions, or instrument completeness. Result producers and verifiers share repository code in places, so independent replication remains a higher evidence tier. The report uses three distinct verbs: **validate** an existing record and gates; **recompute** metrics from committed rows without environment execution; and **rerun** the registered environment or QEMU workload. Commands below are labelled accordingly.

The architecture and post-hoc analysis record Python 3.12.6, NumPy 2.2.6, and PyTorch 2.6.0+cu124 on Windows. requirements-prediction-report-v1_2.txt pins the direct audit dependencies, including PyTorch 2.6.0. The CUDA-enabled wheel was used with CPU inference: four Torch threads for incident dynamics and 16 for the historical table audit and checkpoint restoration. A fresh Python 3.12 virtual environment is recommended. This is a direct-dependency specification, not a cross-platform bitwise lock; another platform or wheel must pass the exact recomputation checks rather than assume equivalence. The broader requirements-research.txt alone is insufficient because it omits PyTorch.

The exact pre-report scientific-evidence snapshot for this review freeze is Git commit c5098bfd099e8ef6c441f42dba90071f3bf71668, which contains the retained v3 and v4 results; the earlier v2 evidence snapshot was f6b45c759830c0148a06b674c980fb9761ac0c8c. Use the revised report's own checkout for the commands below; that earlier commit locates scientific evidence, not the later manuscript, checkpoint archive, or release-checker fixes. The supported clean-source path recomputes the historical architecture tables and post-hoc sensitivity, incident-dynamics inference/calibration/selection/statistics, then checks the report against its retained freeze. The paper checker verifies selected evidence arithmetic and document bindings; it does not independently rerun every underlying study or certify the scientific assumptions.

The prepublication audit found nine of the original 18 matched-study checkpoints missing from the distributed archive. All nine were restored from the original development recipe at 16 Torch CPU threads, with exact checkpoint hashes and validation traces and no final-data access. Seven completed four-thread attempts differed and were rejected; the interrupted four-thread physical-GRU attempt was not admitted. cross_domain_checkpoint_restoration_review_v1_2.json retains that history. All 18 original checkpoints and the byte-identical frozen OS base input are now packaged under artifacts/cross-domain-inference-v1.2/ and bound by cross_domain_inference_archive_v1_2.json.

The separate archived-inference wrapper reconstructs the same already-opened deterministic rows, recomputes all original domain records, common-episode tables, and attribution sensitivity diagnostics, and compares them exactly to the retained results. All eight checks pass. It uses temporary storage for the compressed base input, requires no local target/ data or historical Git objects, and does not write the old results, retrain, select thresholds, open new final seeds, or execute QEMU or the external simulator. This is reproduction of historical evidence, not new confirmatory evidence; the producer and checker share numerical routines.

The older Safety-Gymnasium audits additionally require historical Git objects and a separate locked environment: Python 3.10.20, Safety-Gymnasium 1.0.0, Gymnasium 0.28.1, Gymnasium-Robotics 1.2.2, MuJoCo 2.3.3, NumPy 1.23.5, and pygame 2.1.0. They are not included in the minimal clean-source audit. The umbrella checker reads prior verification records rather than repeating their experiments; several legacy checkers write their default report paths. Inspect their prerequisites and redirect supported outputs before running them against a frozen release.

```
# From the repository root; Python 3.12 in a fresh environment
python -m pip install -r requirements-prediction-report-v1_2.txt
# Read-only audits; no retraining or new final simulator execution
python scripts/verify_cross_domain_archived_inference_v1_2.py
python scripts/verify_incident_dynamics_portable_v2.py
# Regenerates the v3 and v4 catalogs from their seeds and recomputes inference (LightGBM, PyTorch; about 30 CPU-minutes
each)
python scripts/run_incident_dynamics_v3_study.py verify
python scripts/run_incident_dynamics_v4_study.py verify
python -m unittest discover -s scripts -p test_incident_dynamics.py
python -m unittest discover -s scripts -p "test_prediction_report*.py"
python scripts/verify_cross_domain_world_model_paper_v1_2.py
```

The paper checker is read-only by default: it refuses changed release bindings and preserves the retained records on any failed check. UTF-8 text bindings permit only LF/CRLF conversion; PDF, checkpoint, and catalog digests remain byte-exact. For deliberate prepublication authoring only, build with python scripts/build_prediction_is_not_permission_v1_2.py, visually inspect the PDF, then explicitly use the paper checker's --refresh option. Never refresh to make an unexplained mismatch pass. The generic shared builder's defaults target a different report. After publication, changes require a new version rather than refreshing this release.

The attribution verifier recomputes every aggregate from 128 per-seed summaries, checks all four compressed row catalogs, exact seed membership, 10,000 paired resamples, adapter/protocol digests, locked runtime source, zero actuator authority, and non-promotion. The output-ablation v3 verifier performs the analogous retained-catalog recomputation for both arms and 51 provenance, toolchain, authority, and interpretation checks without simulator execution; it validates the historical report blob at the recovery registration commit rather than treating later report versions as experimental drift. The post-hoc verifier independently reconstructs the common-episode rollouts, adjusted intervals, leave-one-out results, and common-proposal scores. The authority inventory records 9/9 neural-protocol and 128/128 physical-runtime tests while distinguishing them from QEMU evidence. The paper verifier recomputes the seven-endpoint and 11-output Bonferroni sensitivities and checks required claim boundaries, PDF metadata, figures, tables, page count, source/PDF digests, and the evidence snapshot.

12. Conclusion

The paper's central contribution remains methodological, with registered prediction results that improve in stages. Incident-motivated development (v2) meets its four error-reduction gates against the earlier paper ensembles. Studies v3 and v4 then improve on the v2-selected models themselves: v4 meets every Physical gate, including the historical guard that v3 failed, and the FerrumOS gain passes on held-out mechanisms but not on the v2 population, where an unannounced drift reversal bounds five-step accuracy. The selected models are boosted per-horizon corrections on the frozen v2 ridge (OS) and on the frozen v2 ensemble (Physical), not a new JEPA objective. The Dream-RSI-inspired curriculum provided no gain, and warning improvements are uneven and carry false-positive costs.

A world model can lead a matched rollout table, respond directionally to interventions, fail at a frozen operational threshold, and still remain correctly subordinate to policy and capability. The earlier horizon-composition-sensitive ranking, zero-coverage latent-hazard estimand, all-stop PyBullet policy, planner-dominated Safety-Gymnasium outcomes, exploratory risk-source comparison, and unfavorable output-value ablation remain intact. Better predictions and warnings in a new synthetic population do not establish that applying those warnings improves outcomes.

The defensible claim is not that FerrumOS or Physical JEPA is safe. It is that predictive evidence, warning quality, effective action changes, realized outcomes, and authority can be measured as separate objects with frozen access boundaries and retained failures. Deterministic blocking is empirically monotone on tested paths, but monotonicity is not benefit; capability, confirmation, revision checks, and actuator denial remain independently necessary. No protected artifact is

promoted.

12.1 Submission scope

The appropriate venue identity is cyber-physical systems, runtime assurance, dependable agents, or evaluation methodology. Cross-domain means the method is exercised separately in an agentic OS and a physical-runtime simulation-not that one model transfers between them. The absence of live hardware and independent execution excludes robotics-deployment claims.

12.2 Remaining evidence tier

The targeted local attribution, its registered post-hoc sensitivity checks, and the retrospectively recovered output-value analysis are complete. The output-ablation design and simulator execution were prospective, but its v1 result failed the final integrity gate; the recovered estimates are therefore not a successful prospective result. Repeating that evidence class would require a newly registered untouched seed range rather than relabelling the recovery. The higher-value evidence-class changes remain independent execution of a frozen controller/shield benchmark, actuator-disabled live HIL with physical clocks and interfaces, and externally controlled labels or assessment. For FerrumOS, independently operated longitudinal use and concurrent preview remain higher value than more synthetic prompts.

12.3 Release rule

Technical Report v1.2 is the prepublication evidence-changing successor to v1.1. Version 1.1 and its repository tag remain intact; v1.2 adds the retained-catalog output-value evidence and the registered incident-dynamics v2, v3 and v4 studies, including the failed v3 rule. The earlier September v1.2 PDF and manuscript remain byte-identifiable historical artifacts. Until v1.2 is published or tagged, review-driven corrections replace the previous candidate and refresh its freeze manifest. Publication or tagging makes that source, PDF, verifier, and manifest immutable; later evidence-changing work must then create a new report version and retain failed frozen results. This local revised release is not an external publication. Promotion requires a separate prospective deployment protocol naming exact targets, rollback, capabilities, post-execution verification, and gates. Publication of this report is not that protocol.

Appendix A. Claim-to-evidence ledger

The incident-motivated addition supports only prediction and warnings: four adjusted primary contrasts and two recall gates pass on a fresh generated population, with all historical point-estimate guards passing. Its complete control table, FPR cost, source-label boundary, failed v1 attempt, and unchanged-authority record are in Section 8.7 and incident_dynamics_result_v2.json. It supplies no additional intervention or outcome claim.

This ledger is part of the report rather than a supplementary marketing summary. It identifies the evidence object that supports each major statement and the closest claim that the same object does not support. The purpose is to make category errors visible during review.

Table A1. Claim-to-evidence ledger.

Claim	Primary evidence	Supported result	Explicit exclusion
Physical JEPA leads matched alternatives	Shared frozen physical final catalog	Lowest H=1, H=3, H=5 rollout error; paired intervals exclude zero	Not universal JEPA superiority
FerrumOS registered ranking differs	Horizon-specific FerrumOS endpoints plus common-episode sensitivity	GRU ensemble leads registered H=1/H=3; JEPA leads all common-episode horizons	Ranking reversal is not isolated from episode composition

Claim	Primary evidence	Supported result	Explicit exclusion
Selected models encode interventions	Paired temporal counterfactual catalogs	100.00% and 93.36% directional accuracy	Not calibrated hazard recall
Shifted scores are weakly calibrated	Frozen reliability, Brier, and ECE outputs	ECE 0.238143 and 0.292887	Not a formal probability guarantee
Learned marginal caution is evaluated	Three-arm 512-case catalogs	Zero marginal blocks at threshold 0.99	No learned safety-value claim
Runtime preview is integrated	FerrumOS QEMU ring-3 shadow path	H=1-H=5 p99 guest time 3 ms	Not production or hard real-time timing
Multiple clients remain correlated and isolated	Four WebSocket clients, 128 responses	No leakage; Jain fairness 0.999937	Not parallel inference or distributed execution
Visible assistant mediation works in bounded sessions	Three QEMU boots, 24 requests	Reads execute; writes await confirmation; deletes block	Not independent users or production telemetry
External physical streams can be replayed	284,398 HAI transitions	Fault-condition error and event diagnostics	Not live Ferrum HIL or physical recovery
Anchor-Lab adds external embodiment data	Six publisher files, 2,925,558 rows	Timing and command/state fields are finite	Not semantically valid for direct v5 scoring
3D geometry/contact stress is exercised	288 local PyBullet DIRECT cases	Contact and simulated recovery are measured	Not practical learned safety at 100% intervention
Controller and shield are jointly evaluated	Safety-Gymnasium final seeds 6000-6127	Planner-only explains most cost reduction; union passes naive-baseline gates; no statistically resolved planner difference in completion or cost	Not independent, sensor-only, physical, or learned-superiority evidence
Risk-source pipelines are compared	Frozen attribution seeds 8000-8127	Exploratory JEPA-only versus full: -25 hazard steps and -0.81 intervention points; adjusted and leave-one-out intervals exclude zero	Not architecture-only causality, preserved completion, or superiority to planner
Observed JEPA output values are intervened on	Prospectively specified experiment; complete retained paired catalogs, seeds 9000-9127	Warning FPR +1.67 and intervention +0.96 points survive seven-endpoint and 11-output Bonferroni checks; precision -15.26 points is pointwise only; completion and hazard intervals include zero	Recovered result is retrospective; not architecture-wide causality, successful prospective evidence, or benefit
Incident-dynamics v3/v4 improve on v2-selected models	v3 and v4 results on fresh populations A and B	v4: Physical passes every gate; OS passes on held-out mechanisms (20.0%/20.8%)	Not a pass of the registered rule (OS population A: 15.6%/5.7%); v3 rule failed; no RSI gain
Deployment stayed unchanged	Recomputed protected SHA-256 digests	Every protected artifact is byte-identical	Not a deployment or release result

A.1 Evidence precedence

When two measurements appear to support different narratives, the operationally closer measurement takes precedence for the operational claim. For example, the physical JEPA's rollout advantage remains valid even though its learned-only intervention count is zero. The former supports dynamics prediction; the latter controls any statement about safety-gate value at threshold 0.99. Likewise, zero union contacts in the 3D run does not supersede 0% task completion and 100% intervention. The complete outcome vector is the result.

A.2 Negative-result taxonomy

The study distinguishes several types of negative evidence. A scientific negative is a completed comparison whose registered estimand does not support the hoped-for effect, as in zero learned marginal caution. A gate pass can still contain a negative marginal contrast, as v14 does when the union improves completion but worsens hazard cost relative to the planner. An engineering negative is a system behavior that is measurable but unusable, as in serial multi-client latency or the all-stop 3D policy. A compatibility negative occurs before model scoring when external data do not share the required semantics, as with Anchor-Lab. A failed result is different: the output-ablation design and simulator execution were prospective and its catalogs completed, but the final protected-file check failed, so only the subsequently registered v2 recovery may report them as retrospective estimates. These categories must not be rewritten as missing data or collapsed into a successful prospective result.

Appendix B. Frozen-gate and artifact audit

The paper inherits the study's frozen-gate discipline. Each verifier checks its own source objects and writes a result that is subsequently bound by the umbrella verification record. The paper verifier does not recompute every experiment; it requires the already committed umbrella result to pass, verifies its non-promotion and protected-digest assertions, and then binds the manuscript and rendered PDF to that evidence snapshot.

Table B1. Frozen-gate and artifact audit by stage.

Stage	Frozen before final access	Verification requirement	Mutation boundary
Architecture selection	Methods, seeds, updates, parameter tolerance, validation rule	Final inaccessible during every selection run	New research model outputs only
Architecture evaluation	Selected checkpoints and final-catalog generator	One final opening; paired bootstrap finite	No deployed artifact path
Learned contribution	Model, calibrator, threshold, three arms	One final opening; honest zero marginal result	No post-final threshold tuning
FerrumOS shadow	Guest image source, command classes, horizons	Correlated responses, runtime fields, source-disk hash	Disposable disk copies only
Multi-client contention	Four-client schedule and correlation IDs	128 responses, isolation, fairness, recovery	Execution RPC unavailable
Natural use	Prompts, sessions, privacy schema	24 bounded rows; forbidden fields absent	Researcher-operated disposable guests
External intake	Source revisions and semantic checklist	File counts, finite rows, compatibility decision	No invalid feature projection
Recorded replay	Projection, faults, descriptive threshold	All transitions finite; authority counters zero	No artifact retraining or actuation
3D stress	Bodies, obstacles, cases, recovery rule	All outcomes and Wilson intervals retained	PyBullet DIRECT; actuator authority zero
External useful-autonomy test	Runtime lock, dev/final seeds, candidates, five arms, joint gates	One untouched final opening; raw union rows and all arms independently recompute	Safety-Gymnasium DIRECT; privileged planner; actuator authority zero
Paired planner-union uncertainty	Seed pairing, estimands, 10,000 resamples, bootstrap seed	Completion and hazard-cost differences independently recompute; both intervals include zero	Post-hoc analysis of committed episode summaries; no final rerun
Registered sensitivity v1	Common episodes/proposals, three-comparison family, 128 omissions	Exact recomputation, adjusted intervals, and source-arm reproduction pass	No simulator, retraining, threshold change, or promotion

Stage	Frozen before final access	Verification requirement	Mutation boundary
Output-ablation v1 execution	Two feature interventions, seeds 9000-9127, estimands, and bootstrap seeds fixed pre-catalog	Both prospectively executed catalogs complete; final protected-report digest check fails	Prospective result ineligible; failure retained
Output-ablation recovery v2 / verification v3	Retained catalog digests and unchanged estimands	51/51 checks pass; aggregates, scores, features, seeds, and paired effects recompute	Retrospective reportable analysis only; no simulator rerun, result reclassification, or promotion
Paper freeze	Required claims, boundaries, figures, metadata	Text, hashes, pages, evidence snapshot pass	Documentation artifacts only

B.1 Protocol and amendment chronology

Hosted Git history timestamps the registered files after commit; it does not prove absence of any earlier equivalent observation. The table is an audit index, not a claim that every stage was scientifically successful.

Table B2. Protocol and amendment chronology.

Stage	Registration commit / time (IST)	Frozen change	Outcome
Safety-Gym v1-v4	383e525-f9d1a52, 30 Aug 17:32-19:24	Initial recovery, new final ranges, realized-cost gate	v1 selection failed; v3 final failed; v4 retained mixed record
Safety-Gym v5-v8	5a2a07c-2ae1502, 30-31 Aug	Earlier warning, bounded steering, braking, adaptive exit	Development failures; no promoted final
Safety-Gym v9-v13	22f9774-4f62544, 31 Aug	Privileged planner factorization, tangent correction, recall repair, Simplex trial	v10/v12 finals failed; v13 failed development
Safety-Gym v14	bcca0b6, 1 Sep 23:21	Synchronized 20-step nominal oracle and corrected cost accounting	One final pass against naive-baseline gates; planner-relative tradeoff retained
Attribution v1	df58e72, 5 Sep	Four fixed risk sources, seeds 7000-7127	Schema failure after two stages; partial catalog retained and excluded
Attribution v2	e954a2c, 5 Sep	Threshold alias only; new seeds 8000-8127	Completed once; all evidence checks pass; no promotion
Post-hoc sensitivity v1	cd5e9b6-da32691, 8 Sep	Common episodes/proposals, multiplicity, paired directions, leave-one-out	Exact recomputation passes; no simulator, retraining, or promotion
Output ablation v1	6f70724, 8 Sep	Observed frozen JEPA values versus preregistered development-mean masking; estimands, seeds 9000-9127, and bootstrap seeds fixed pre-catalog	Both catalogs completed prospectively; protected-report digest check failed; prospective result ineligible
Output recovery v2	c47b8ba, 8 Sep	Analysis-only reuse of complete retained paired catalogs	Retrospective estimates written; no simulator rerun or prospective reclassification
Output verification v3	9cd2c0a-6a6da0a, 8 Sep	Portable text-digest rule and historical report-blob validation	51/51 checks pass in current and clean Windows checkouts; evidence values unchanged
Incident-dynamics v1	63a1059-fd036c3, 18 Oct	New curriculum, validation-selected models, independent calibration, frozen primary gates	Historical duplicate-step schema rejection before final model scoring; failed attempt retained
Incident-dynamics v2	3086e52-645fd8f, 18 Oct	General duplicate-step exclusion, unchanged models and thresholds, fresh seed 281014	All four prediction and both warning gates pass; six historical regression guards pass
Incident portable audit	dc8d9dd-f6b45c7, 18 Oct	Retained-catalog inference, calibration, selection and metric recomputation	48/48 checks match in a clean source archive; only JSON newline encoding may differ

Stage	Registration commit / time (IST)	Frozen change	Outcome
Incident-dynamics v3 plan	d68a1c1-2b76913, 18 Oct 22:50-23:45	24 verified sources, ten families per domain with three held out, candidate families, guarded generator	Development only; no final access
Incident-dynamics v3 protocol	c574edb, 19 Oct 01:37	Populations A/B, three-block selection metric, 20% gates, 5% guard, RSI claim rule	Registered before selection
Incident-dynamics v3 selection	7e51a28, 19 Oct 02:12	Direct boosted OS and one-step boosted Physical ensembles; k=0 warnings	Frozen before final access
Incident-dynamics v3 final	366dee6, 19 Oct 02:24	One opening of seeds 391014 and 391015	Primary rule failed (six effect sizes, one guard); 48/48 verification checks pass
Incident-dynamics v4 protocol	c9418e0, 19 Oct 03:13	Physical direct family and committee; unchanged gates; fresh seeds	Registered after the v3 failure
Incident-dynamics v4 selection and final	4b66f7c-c5098bf, 19 Oct 03:27-03:34	Direct Physical ensemble; FerrumOS carried forward	Physical passes all gates; OS population A fails; 46/46 checks pass

B.2 Protected deployment inventory

The umbrella record protects four deployed objects: the FerrumOS encoder, FerrumOS transition, FerrumOS manifest, and Physical JEPA transition. Their observed digests equal their registered expected digests. Architecture checkpoints, adapters, catalogs, replay projections, and simulator results remain research artifacts. A passing verifier cannot change that status.

B.3 Reproduction order

Start with the read-only commands and explicit availability limits in Section 11. The archived-inference wrapper reproduces the old architecture and sensitivity records from the completed checkpoint archive; the incident-dynamics portable verifier recomputes its separate evidence; the paper verifier checks the retained release without rewriting its freeze. For runtime studies, inspect selection-access logs, final-open accounting, no-execution records, and actuator-authority counters as retained evidence. The separate output-ablation v3 verifier needs its historical Git blob and locked runtime. A prior verifier's recorded pass is not a fresh independent execution, and the paper checker's bounded assertions are not a general proof of scientific validity.

B.4 Archival boundary

Technical Report v1.2 is intended to be archived together with its manuscript, figures, paper verification record, and freeze manifest while v1.1 remains separately retrievable. Dataset or software records may receive their own archival identifiers because they are independently reusable objects. Reciprocal links should identify the exact immutable report version and the exact dataset or software version without implying that a repository's moving branch is itself frozen. The DOI fields are intentionally not invented in this pre-archive build; they can be added only after the archival service reserves or publishes the corresponding identifiers.

Appendix C. Artifact locator

The October addition is rooted at docs/research/incident_dynamics_protocol_v1.json and its v2 amendment. The adjacent incident_dynamics_selection_v2.json, incident_dynamics_result_v2.json, incident_dynamics_verification_v2.json, and incident_dynamics_portable_verification_v2.json bind selected weights, calibration, results, and the 48-check audit. incident_dynamics_sources_v1.json records all source URLs and representation limits. docs/research/artifacts/incident-dynamics-v1/ retains development traces, checkpoints, validation/calibration catalogs, the earlier report, and the failed final attempt; incident-dynamics-v2/final/ contains the successful retained final catalogs and predictions. Run scripts/verify_incident_dynamics_portable_v2.py to reproduce inference and statistics without retraining or generating another final catalog.

The repository is the executable supplement to the narrative. Table C1 lists the shortest path from a headline claim to its primary machine-readable evidence. Verification files are intentionally separate from result files so that a result producer is not the sole authority for its own gate.

Table C1. Artifact locator from headline claim to machine-readable evidence.

Evidence object	Repository path	Review use
Prospective cross-domain protocol	docs/research/cross_domain_world_model_improvement_protocol_v1.json	Registered methods, seeds, budgets, access and promotion rules
Architecture selection	docs/research/cross_domain_world_model_selection_v1.json	Check chosen family and denied final access
Architecture result	docs/research/cross_domain_world_model_architecture_result_v1.json	Recompute H=1, H=3, H=5 tables and paired intervals
Learned-contribution result	docs/research/cross_domain_learned_contribution_result_v1.json	Recompute three-arm confusion matrices and marginal counts
FerrumOS shadow result	docs/research/world_model_v3_4_shadow_runtime_v1.json	Inspect guest timing, loaded artifact and no-execution fields
Multi-client result	docs/research/world_model_multiclient_contention_result_v1.json	Inspect correlation, fairness, leakage and recovery
Natural-use result	docs/research/world_model_natural_use_result_v1.json	Inspect bounded sessions and privacy assertions
External intake result	docs/research/world_model_external_case_intake_result_v1.json	Inspect source revisions, rows and compatibility decision
Recorded replay result	docs/research/physical_jepa_recorded_hil_replay_result_v1.json	Recompute fault-condition and authority-disabled metrics
3D stress result	docs/research/physical_jepa_multi_embodiment_3d_result_v1.json	Recompute completion, intervention, contact and recovery
External useful-autonomy protocol	docs/research/physical_jepa_safety_gymnasium_protocol_v14.json	Inspect runtime lock, seed boundary, candidate policy and frozen joint gates
External useful-autonomy result	docs/research/physical_jepa_safety_gymnasium_result_v14.json	Recompute five arms, planner divergence, learned-only attribution and realized-cost reduction

Evidence object	Repository path	Review use
External useful-autonomy verification	docs/research/physical_jepa_safety_gymnasium_verification_v14.json	Confirm raw cases, exact seeds, hashes, gates, authority zero and non-promotion
Paired planner-union uncertainty	docs/research/physical_jepa_safety_gymnasium_paired_uncertainty_result_v1.json	Recompute seed-matched completion and realized hazard-cost difference intervals
Risk-source attribution protocol	docs/research/physical_jepa_safety_gymnasium_attribution_protocol_v2.json	Inspect fixed factors, thresholds, recovery, and fresh seed boundary
Risk-source attribution result	docs/research/physical_jepa_safety_gymnasium_attribution_result_v2.json	Recompute four pipeline arms, episode statistics, and paired differences
Risk-source attribution verification	docs/research/physical_jepa_safety_gymnasium_attribution_verification_v2.json	Confirm row catalogs, seed sets, digests, runtime, authority, and non-promotion
Registered post-hoc sensitivity	docs/research/cross_domain_world_model_posthoc_sensitivity_result_v1.json	Common episodes, common proposals, adjusted intervals, paired directions, and leave-one-out checks
Post-hoc sensitivity verification	docs/research/cross_domain_world_model_posthoc_sensitivity_verification_v1.json	Exact independent recomputation without simulator execution
Output-ablation failed attempt	docs/research/physical_jepa_safety_gymnasium_output_ablation_failed_attempt_v1.json	Confirm prospective v1 failure, complete catalog retention, and result ineligibility
Output-ablation recovery protocol	docs/research/physical_jepa_safety_gymnasium_output_ablation_recovery_protocol_v2.json	Inspect the subsequently registered analysis-only recovery boundary
Output-ablation recovery result	docs/research/physical_jepa_safety_gymnasium_output_ablation_recovery_result_v2.json	Recompute observed-versus-masked aggregates and paired intervals from retained catalogs
Output-ablation verification amendment	docs/research/physical_jepa_safety_gymnasium_output_ablation_verification_amendment_v3.json	Inspect the portable digest and historical report-blob verification policy
Output-ablation recovery verification	docs/research/physical_jepa_safety_gymnasium_output_ablation_recovery_verification_v3.json	Confirm 51/51 checks, failed-v1 interpretation, authority zero, and non-promotion
Authority test inventory	docs/research/cross_domain_authority_test_inventory_v1.json	Map component, test class, committed pass, and execution availability
Umbrella verification	docs/research/cross_domain_world_model_improvement_verification_v1.json	Confirm subordinate passes, claim boundaries and protected hashes

Evidence object	Repository path	Review use
Narrative study	docs/research/CROSS_DOMAIN_WORLD_MODEL_IMPROVEMENT_STUDY.md	Read the compact evidence-first study before this full report
Incident-dynamics v3 sources	docs/research/incident_dynamics_sources_v3.json	Inspect 24 source URLs, verification notes, family mapping and caveats
Incident-dynamics v3 protocol and plan	docs/research/incident_dynamics_protocol_v3.json, incident_dynamics_v3_development_plan.json	Inspect populations, gates, holdout rule, disclosed probes and amendments
Incident-dynamics v3 selection	docs/research/incident_dynamics_selection_v3.json	Recompute candidate scores; confirm compact archives and k rule
Incident-dynamics v3 result	docs/research/incident_dynamics_result_v3.json	Read primary endpoints, warnings, RSI ablation and historical guards
Incident-dynamics v3 verification	docs/research/incident_dynamics_verification_v3.json	Confirm 48 regeneration, inference and statistics checks
Incident-dynamics v4 records	docs/research/incident_dynamics_protocol_v4.json, incident_dynamics_selection_v4.json, incident_dynamics_result_v4.json, incident_dynamics_verification_v4.json	Inspect the successor design, its fresh results, and 46 checks

C.1 Recommended audit path

Begin with the prospective protocol and umbrella verification. Confirm promotion_eligible=false, all protected unchanged fields, and the claim-boundary list. Then inspect the architecture and learned-contribution results together: this prevents a favorable rollout table from being detached from the operational zero-result. Continue to the runtime and physical records only for the integration claims they support. In the Safety-Gymnasium evidence, compare all five arms and planner divergence before attributing any outcome to the learned branch. For the output ablation, read the failed v1 record before the recovery result and require the v3 verification interpretation to remain retrospective. Finally, execute the v1.2 paper verifier and compare its recorded manuscript and PDF digests with the archived files.

C.2 Versioning rule

The report version identifies a fixed document, not a promise that every moving repository file will remain identical forever. Version 1.2 is evidence-changing because it binds the retained-catalog output ablation and incident-motivated study; it does not mutate or supersede the evidentiary status of v1.1. Once published or tagged, corrections that alter prose, evidence binding, or layout should create a new immutable report version and retain the old version in archival history. Dataset and software deposits should use their own versioned records when they are independently reusable. A concept DOI may identify the evolving record, while the paper should cite the immutable version DOI used during review.

Appendix D. Seed-level architecture results

Each cell reports H=1/H=3/H=5 normalized endpoint error for one independently initialized checkpoint. Table 3 instead reports error after averaging the three state predictions. These seed results show training variability; the paper's episode bootstrap does not resample this training process.

Table D1. Seed-level architecture results (H=1 / H=3 / H=5 normalized endpoint error).

Domain	Method	Seed 17	Seed 43	Seed 101
FerrumOS	Direct MLP	0.006589 / 0.011355 / 0.010724	0.006766 / 0.010380 / 0.008943	0.005904 / 0.009543 / 0.008113
FerrumOS	Action-conditioned JEPA	0.011977 / 0.014114 / 0.006823	0.011571 / 0.015009 / 0.006900	0.012893 / 0.015745 / 0.007215
FerrumOS	GRU dynamics	0.006254 / 0.009522 / 0.008289	0.006649 / 0.010079 / 0.009221	0.006380 / 0.009540 / 0.007955
Physical	Direct MLP	0.008923 / 0.022246 / 0.035243	0.008352 / 0.019819 / 0.031014	0.007903 / 0.018208 / 0.028402
Physical	Action-conditioned JEPA	0.002967 / 0.007682 / 0.011869	0.003056 / 0.008098 / 0.012635	0.003017 / 0.007989 / 0.012481
Physical	GRU dynamics	0.013441 / 0.032976 / 0.049582	0.012091 / 0.028457 / 0.041401	0.012486 / 0.032275 / 0.050767

D.1 Incident-dynamics v2 development trials

The table retains all 13 completed neural trials. Scores are individual checkpoints' mean validation H=1/3/5 MAE, not test performance. Configuration selection uses the ensemble prediction score, not the average score in this table and not the best individual seed. OS ridge separately searches regularization 0.001, 0.1, and 10 and wins the configuration-level selection; its fit is deterministic. Initial physical controls are retained even when later wider controls improve on them.

Table D2. Incident-dynamics v2 development trials (individual checkpoint validation scores).

Domain / model / width	Seed	Parameters	Completed / selected updates	Individual validation score
OS JEPA / 211	17	99,748	6000 / 5750	0.0019481
OS causal residual / 192	17	122,160	6000 / 5500	0.0015843
OS causal residual / 192	43	122,160	6000 / 5750	0.0015458
OS causal residual / 192	101	122,160	6000 / 6000	0.0015846
OS MLP / 242	17	99,800	6000 / 5250	0.0018875
OS action residual / 64	17	152,304	12000 / 11500	0.0014460
Physical JEPA / 372	17	48,476	6000 / 6000	0.0120193
Physical causal residual / 192	17	56,576	6000 / 6000	0.0110019
Physical JEPA / 777	17	99,911	12000 / 11500	0.0098052
Physical MLP / 900	17	99,932	12000 / 11500	0.0113965
Physical action residual / 192	17	73,840	12000 / 12000	0.0056684
Physical action residual / 192	43	73,840	12000 / 11250	0.0058359
Physical action residual / 192	101	73,840	12000 / 12000	0.0059011

The selected physical ensemble's validation score is 0.0053222; OS ridge's is 0.0013137. These validation scores combine old and added development episodes and should not be compared as if they came from the fresh final population. Selected-model versus same-data control differences also combine ensemble-size and budget effects. All trial settings and checkpoint traces are retained under the artifact locator in Appendix C.

D.2 Incident-dynamics v3 and v4 development trials

Every retained configuration is listed with its seeds and its mean single-seed development score under the registered three-block metric (v2-selected = 1). Ensemble selection scores differ slightly from these means. Crashed, stopped, and never-run trials are recorded in the development ledger and infrastructure_failures.jsonl.

Table D3. Retained incident-dynamics v3 and v4 development configurations. H is history length; Score is the mean single-seed three-block score (v2-selected = 1), with per-block ratios.

Domain	Configuration	Family	H	Weighting	Seeds	Score	Legacy	Incident	Mechanism
OS	o01-ar-mech	AR net	8	uniform	17	1.058	0.958	1.395	0.820
OS	o02-ar-mech-iw4	AR net	8	incident x4	17	0.940	0.955	1.143	0.722
OS	o09-boost-mech	net on v2	8	uniform	17	1.375	1.015	2.074	1.036
OS	o12-boost-bb	net on v2	8	balanced	17	1.000	1.000	1.000	1.000
OS	o16-edd-bb	effect-drift net	8	balanced	17	1.048	1.054	1.328	0.763
OS	o20-gbr-h8-iw8	trees on v2, one-step	8	incident x8	17, 43, 101	0.791	1.028	0.799	0.545
OS	o21-gbr-h3-iw8	trees on v2, one-step	3	incident x8	17	0.792	1.028	0.776	0.571
OS	o22-gbr-h3-iw16	trees on v2, one-step	3	incident x16	17	0.787	1.029	0.765	0.568
OS	o23-gbr-h8-iw8-t1500	trees on v2, one-step	8	incident x8	17	0.825	1.028	0.884	0.563
OS	o24-gbr-h8-iw8-l127	trees on v2, one-step	8	incident x8	17	0.827	1.023	0.890	0.568
OS	o26-gbrd-h8-iw8	trees on v2, direct	8	incident x8	17, 43, 101	0.693	1.001	0.588	0.491
OS	o27-rsi-gbrd-h8-iw8	trees on v2, direct	8	incident x8, RSI	17, 43, 101	0.698	1.001	0.602	0.489
OS	o28-gbrd-h8-iw8-scaled	trees on v2, direct	8	incident x8, + scale	17, 43, 101	0.693	1.001	0.588	0.491
Physical	p01-ar-base	AR net	8	uniform, no mech.	17	0.902	0.921	0.860	0.926
Physical	p13-ar-bb	AR net	8	balanced	17	0.948	1.192	0.855	0.797
Physical	p17-boost-bb	net on v2	8	balanced	17	0.907	1.039	0.890	0.792
Physical	p21-gbr-h8-iw4	trees on v2, one-step	8	incident x4	17, 43, 101	0.660	0.653	0.697	0.631
Physical	p23-ar-mech-u20k-h384	AR net	8	uniform	17	0.781	0.856	0.752	0.735
Physical	p24-gbr-h8-iw4-scaled	trees on v2, one-step	8	incident x4, + scale	17, 43, 101	0.660	0.653	0.697	0.631
Physical	p30-gbrd-h8-iw4-t300	trees on v2, direct	8	incident x4	17, 43, 101	0.584	0.554	0.630	0.568

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